Modeling a Real-World Evacuation Scenario in Mariposa, CA

Presented by

Qijian Gan
(PATH/University of California, Berkeley)

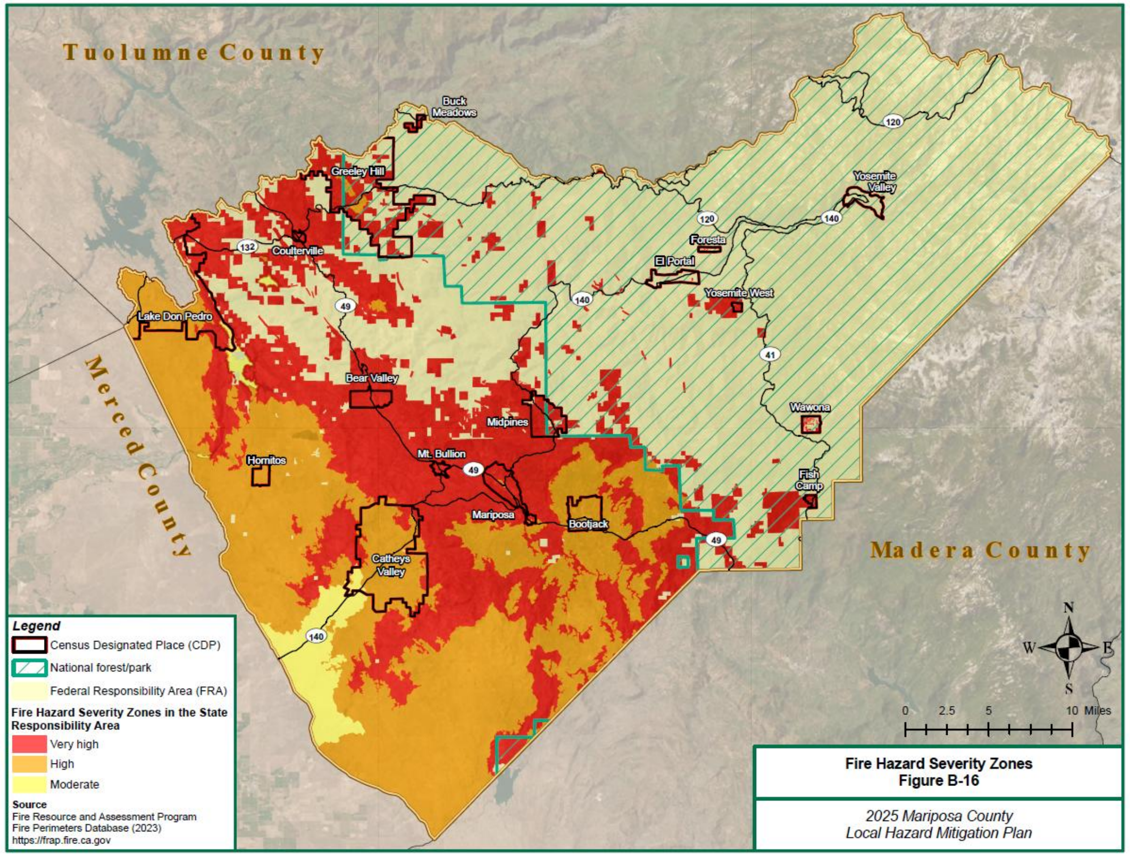
Why Develop a Real-World Evacuation Scenario?

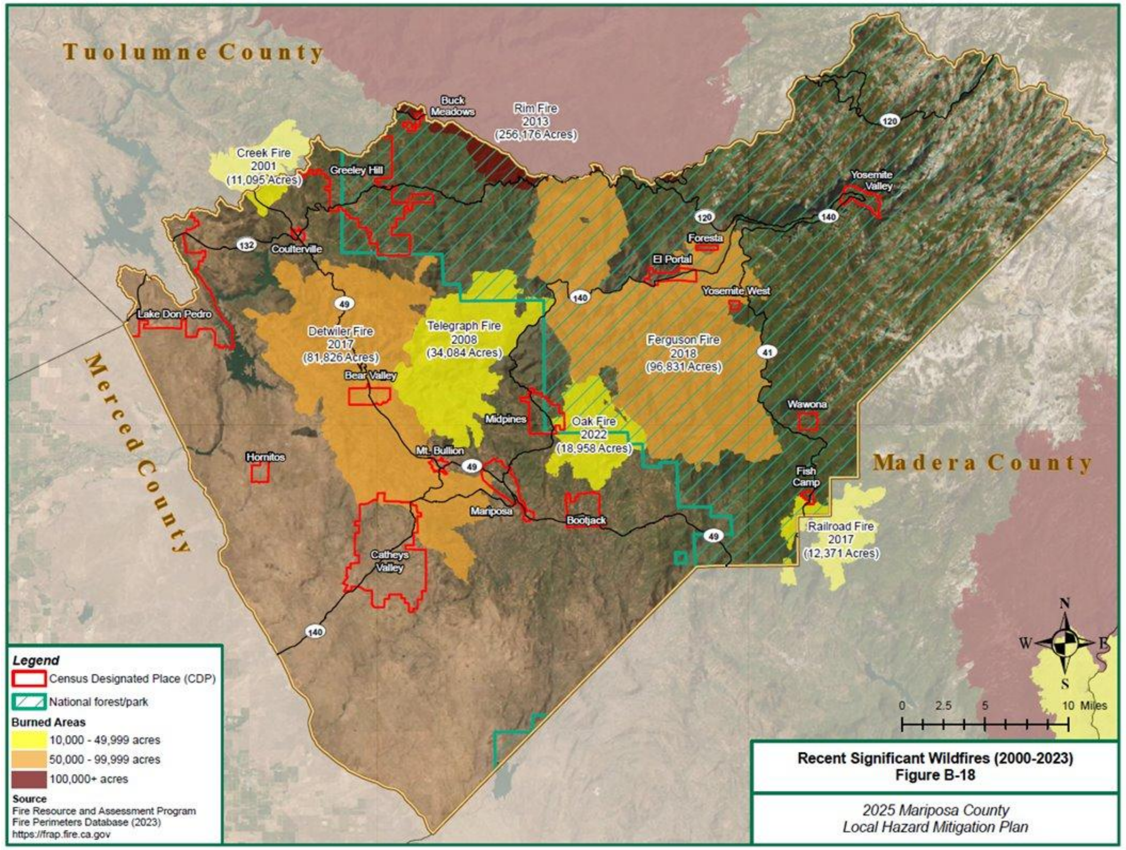
- State-of-the-art studies on the placement of charging stations and evacuation plans typically rely on macroscopic-level planning and/or optimization tools.
  - We want to assess their performance in a high-granularity model with more practical settings.
    - ✓ Placement of fixed and/or mobile charging stations
    - ✓ EV evacuation strategies/plans, especially with Mobile Charging Stations (MCSs)

- We want to conduct the study that has great potential to be deployed at scale and support our community partners' planning and operations needs.

Why Mariposa is Chosen?

It is an area with a very high fire hazard severity!





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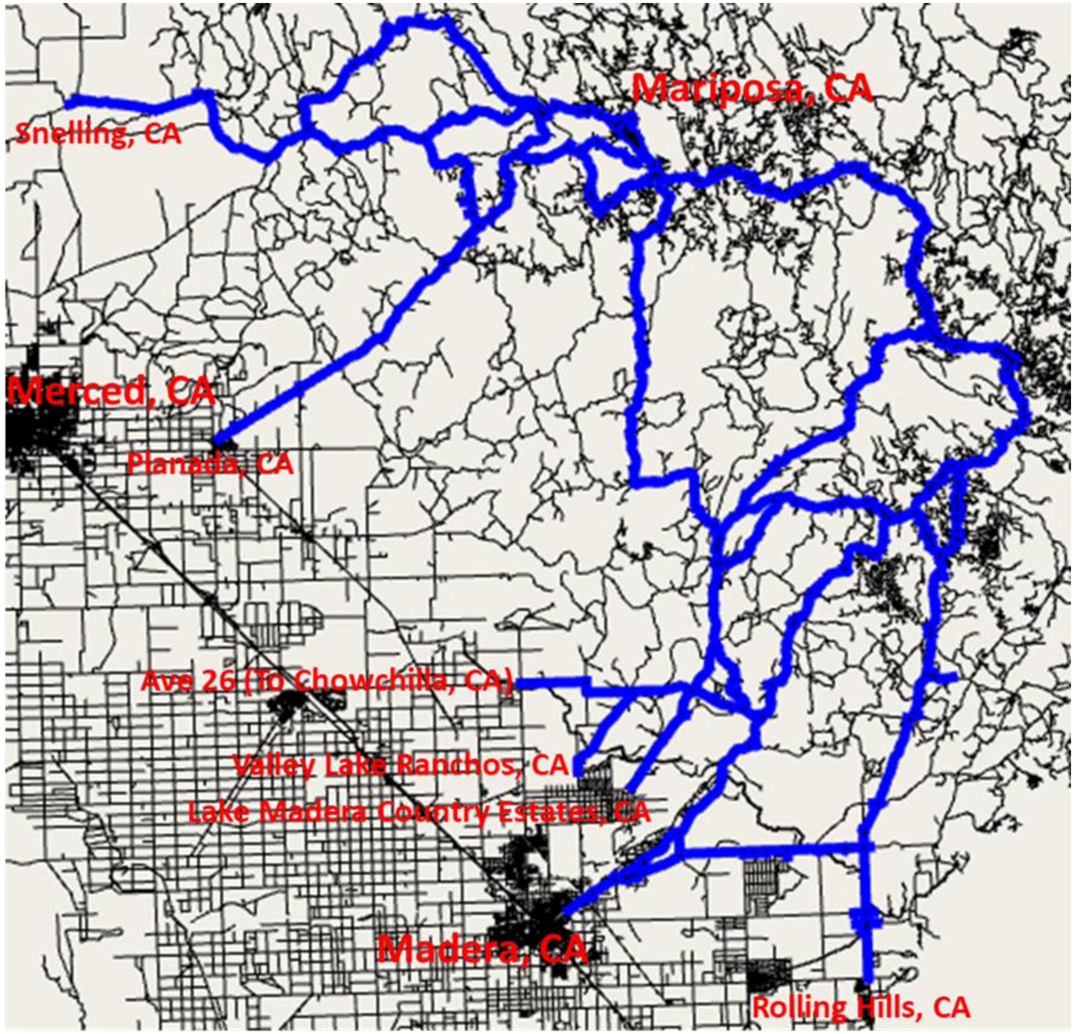
The Mariposa Network in Aimsun



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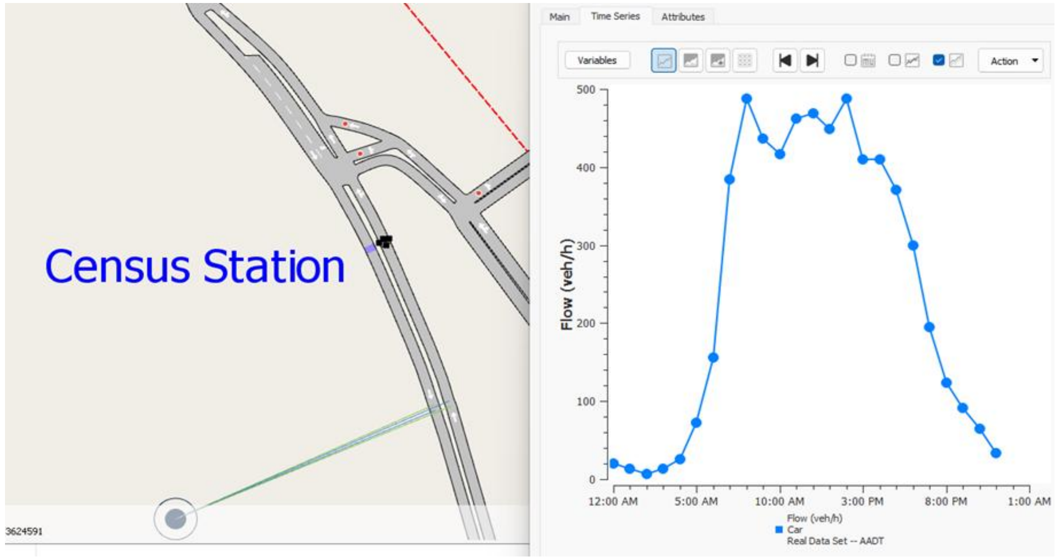
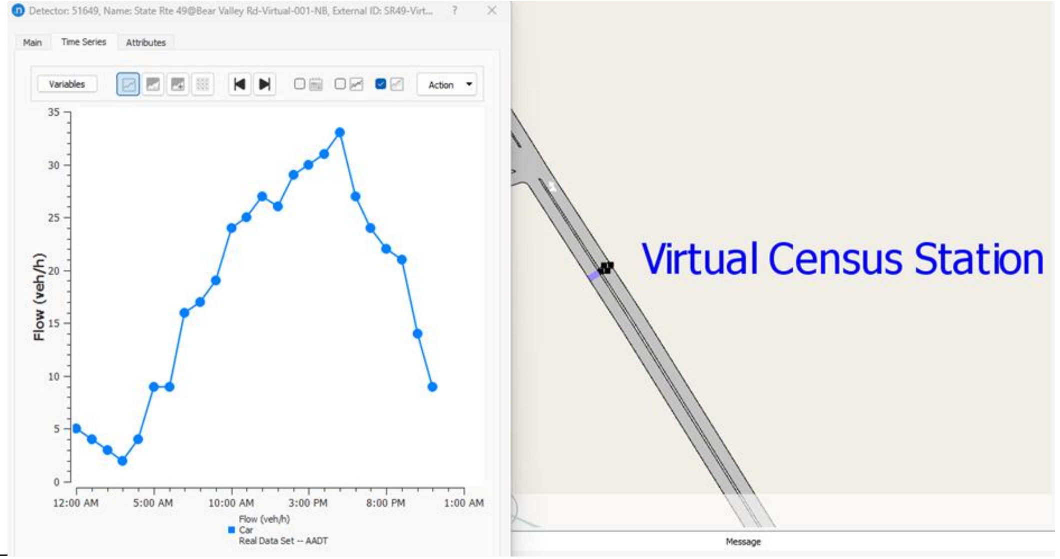
Model Calibration (1/2)

- Data Collection
  - We did not use Vehicle Detector Station (VDS) data for model calibration since there are only 6 VDSs at 3 locations in the study network.
  - We developed algorithms to fuse the following two data sources to generate a set of time-of-day traffic data for model calibration.
    - ✓ Time-of-day traffic data from 16 census stations in PeMS (https://pems.dot.ca.gov/).
    - ✓ AADT data in *Caltrans GIS Data: Annual Average Daily Traffic* at 48 locations.





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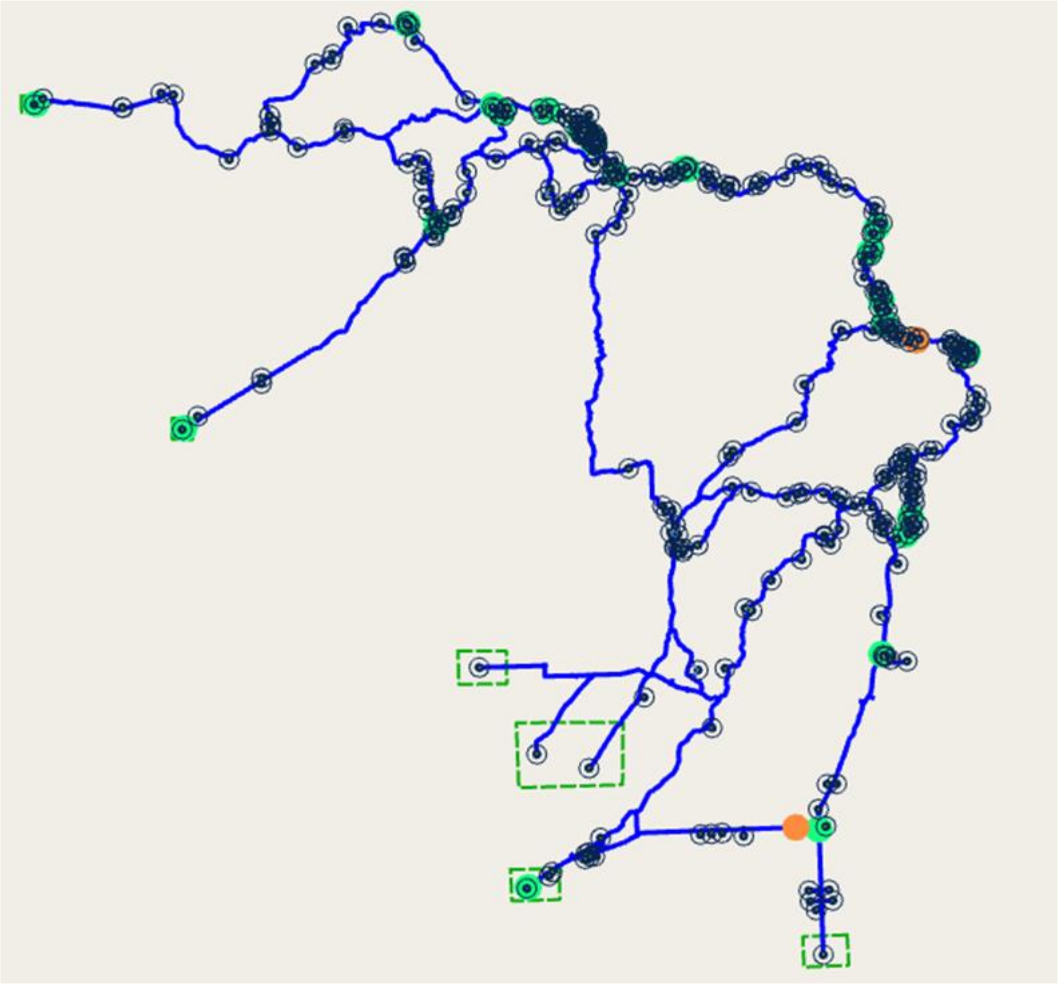
Model Calibration (2/2)

- The entire model has been calibrated for a typical day, including 24 one-hour traffic demand profiles.
- The GEH statistics for the 64 traffic detection points meet the calibration requirements for microsimulation.
  - At least 85% of detection points have GEH less than 5

$$GEH = \sqrt{\frac{2(M - C)^2}{M + C}}$$

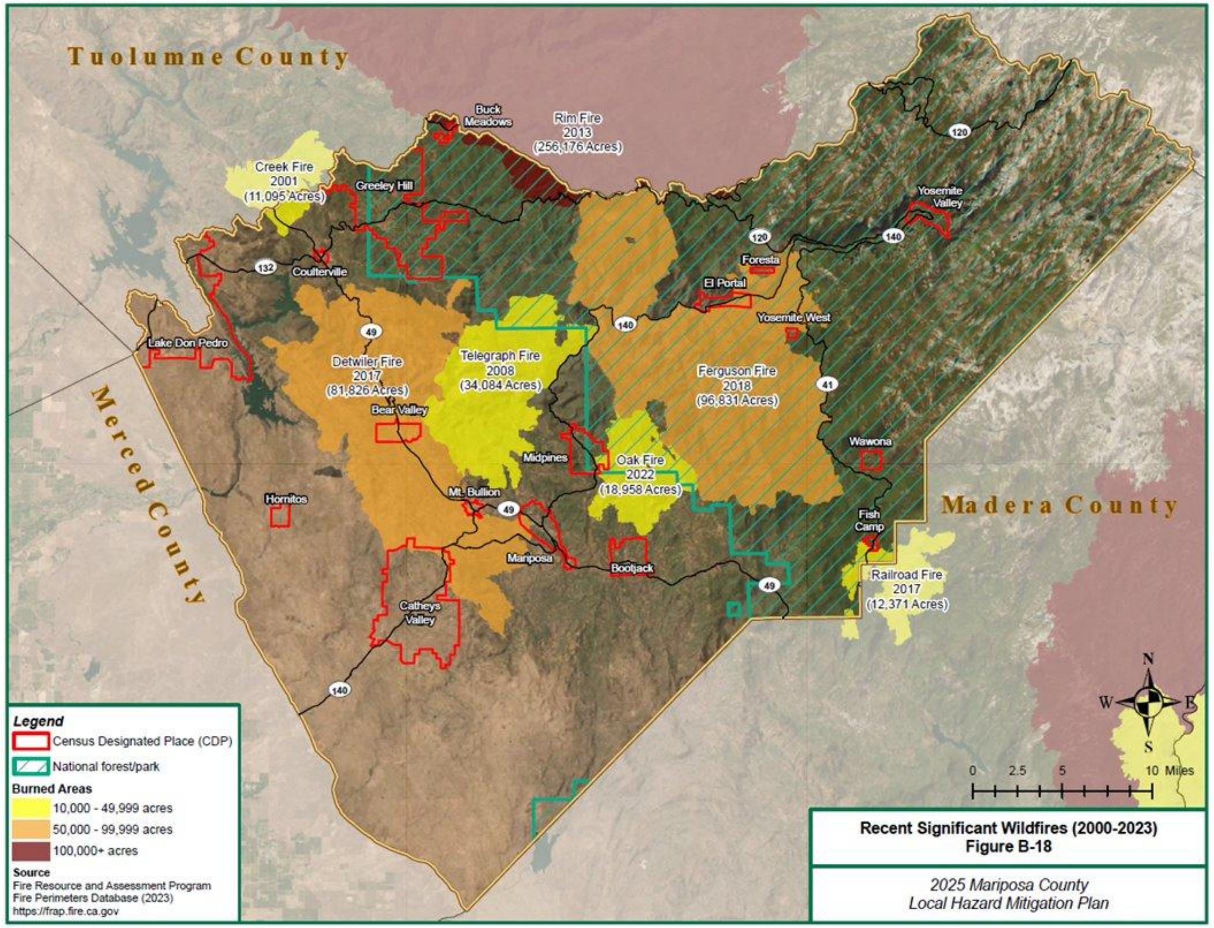
Where M is the hourly traffic volume from the traffic model (or new count) and C is the real-world hourly traffic count (or the old count)

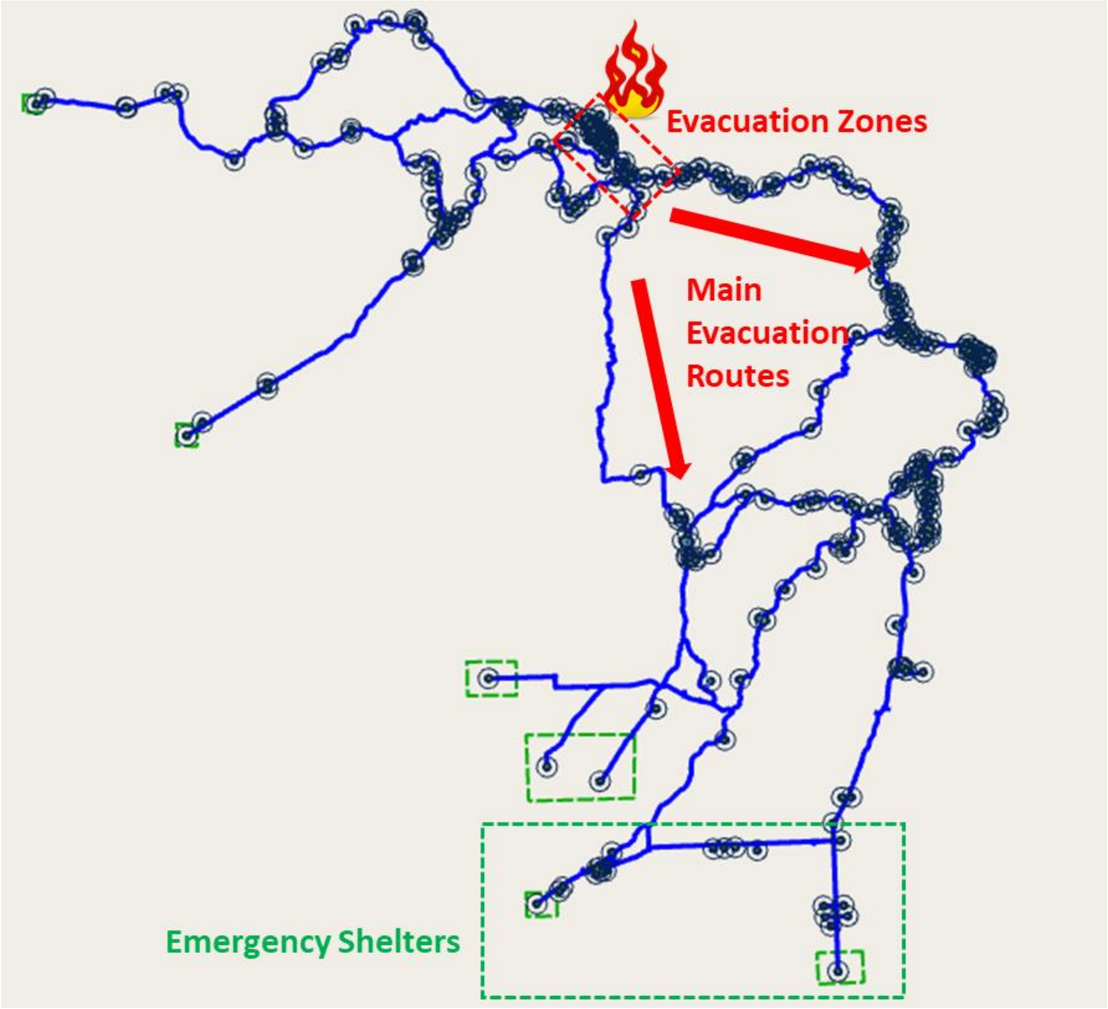


GEH Summary

Number of Objects: 64 | Number of Objects with GEH < 5: 58 (90.63%) | Number of Objects with GEH < 10: 64 (100.00%)

Evacuation Scenario Design (1/7)





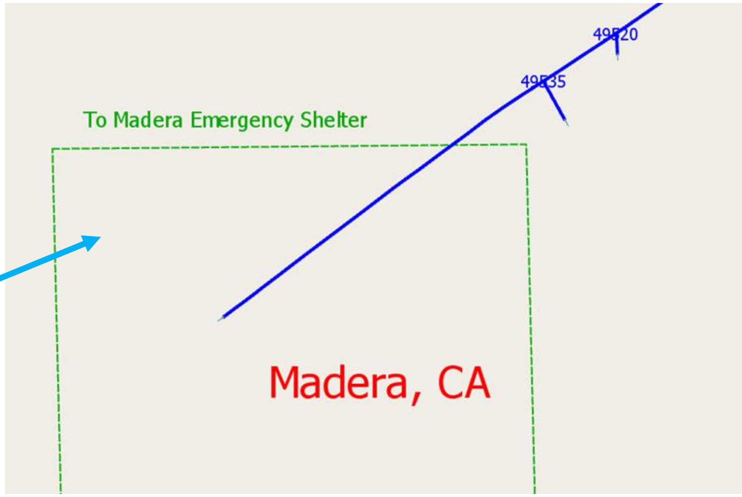
Evacuation Scenario Design (2/7)

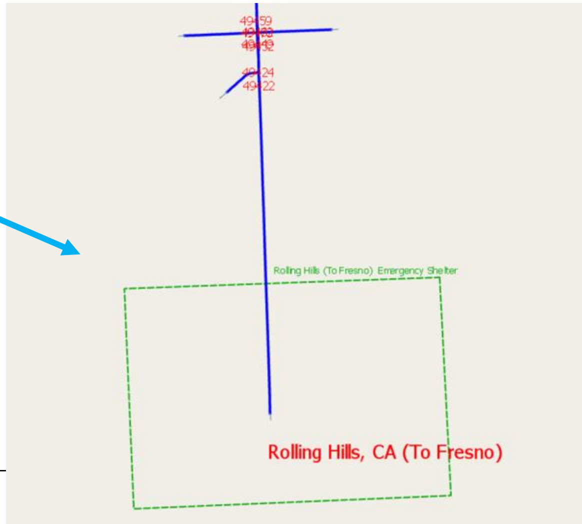
Three Evacuation Zones

About 50 miles

About 55 miles

Two Emergency Shelters



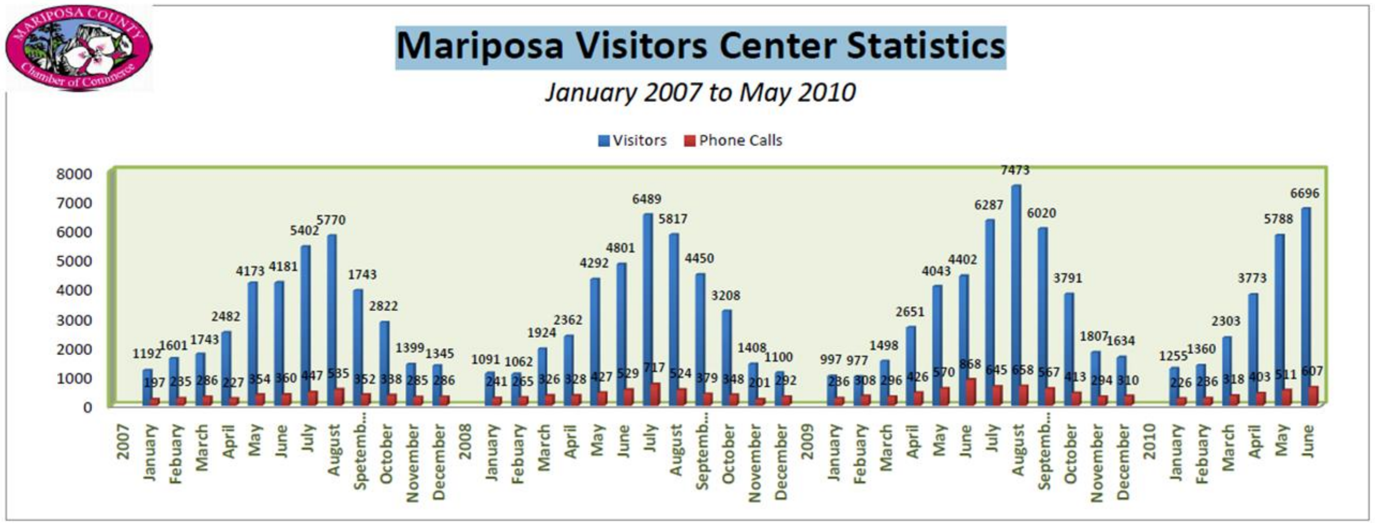


Evacuation Scenario Design (3/7)

Table 3-1: Communities Recognized as Census Designated Places in Mariposa County

Name	Square Miles	Acres	2020 Census Population
Bear Valley	2.42	1,547.28	157
Bootjack	3.85	2,464.32	662
Buck Meadows	0.51	329.07	21
Catheys Valley	23.65	15,135.57	829
Coulterville	0.39	250.53	116
El Portal	1.99	1,274.78	29
Fish Camp	0.40	255.28	373
Foresta	0.37	236.70	49
Greeley Hill	23.98	15,346.57	930
Hornitos	1.14	730.84	38
Lake Don Pedro	19.68	12,593.60	1,769
Mariposa (County seat)	4.06	2,598.37	1,535
Midpines	4.33	2,768.55	382
Mt. Bullion	0.60	385.43	155
Wawona	1.07	687.38	111

Evacuation Scenario Design (4/7)



Estimated Averaged Number of Visitors Staying in the City of Mariposa
= Monthly Visitor Number / 30 * Averaged Length of Stay

Month	Monthly Visitor Number	Averaged Length of Stay	Averaged Daily Visitors Staying in Mariposa
January	1093	3	109
February	1213	3	121
March	1722	3	172
April	2498	3	250
May	4169	3	417
June	4461	3	446
July	6059	3	606
August	6353	3	635
September	4738	3	474
October	3274	3	327
November	1538	3	154
December	1360	3	136

Evacuation Scenario Design (5/7)

34 Fixed & Mobile Charging Stations

ID	Name	Max # of EVs	Fixed Charging Station: DCFC # of Chargers	DCFC Charger Power (kw)	L2 # of Chargers	L2 Charger Power (kw)	Max # of Mobile Charging Stations
50787	Chargepoint & Electrify America Charging Stations	70	12	150	2	6.5	10
50807	Chukchansi Crossing Fuel Station & Travel Center	25			4	12	4
51078	MACT MARIPOSA Charging Station	50			2	6.5	10
51094	Pioneer Market - Tesla Supercharger	50	12	250			8
51107	Yosemite Plaisance Bed & Breakfast - Tesla Destination	5			2	16	1
51135	Charging Stations@Oakhurst Library@Best Western Plus@Oakhurst Supercharger	70	8	250	8	8	10
51145	Chateau du Sureau - Tesla Destination	30			2	16	6
51172	Charging Station @ Chuckchansi Resort & Casino	70			22	10	10
51183	Blink Charging Station @ Chawanakee Unified School District	20	2	120			4
50695	Candidate MCS @Oasis	5					1
50725	Candidate MCS @Mariposa Co. Fire Station 25 Mt. Bullion/Airport	5					1
50734	Candidate MCS @Church of St. Catherine of Siena	10					2
50753	Candidate MCS @California Historical Landmark No. 331	5					1
50778	Candidate MCS @ Wasuma Elementary School & Freson Flats Comm Day School	20					4
50797	Candidate MCS @ Madera County Fire Station 15	10					2
51193	Candidate MCS @ California State Mining & Mineral Museum	30					6
51217	Candidate MCS @ Lutheran Church - Mariposa	20					4
							4
							4
							1
							1
							4
							4
							2
							2
							1
							1
51388	Candidate MCS @Cathey's Valley Park	20					4
51398	Candidate MCS @Valero	20					4
51408	Candidate MCS @Sierra Foothill Charter School	10					2
51422	Candidate MCS @CAL FIRE MMU Catheys Valley Fire Station	5					1
51434	Candidate MCS @Forestry & Fire Protection	10					2
51460	Candidate MCS @CAL FIRE MMU Hornitos Fire Station	5					1
51504	Candidate MCS @Anglers Edge Market	10					2

Evacuation Scenario Design (6/7)

- ❑ Capacity (maximum number) of EVs : 30
- ❑ Fixed Charging Stations
  - ➢ DCFC: 0
  - ➢ L2: 0
- ❑ Mobile Charging Stations
  - ➢ Maximum number of MCSs: 6

Evacuation Scenario Design (4/7)

- Evacuation time period
  - 6 hours from 6 AM to 12 PM
- Vehicles to be evacuated: 2100 vehicles
  - 600 vehicles for visitors
  - 1500 vehicles for the residents
- 50% of the evacuation traffic will be routed to Emergency Shelter #1, and 50% to Emergency Shelter #2.

Evacuation Demand	Centroids	Time Period											
		0000-0600			0600-1200			1200-1800			1800-2400		
		Total # of Vehicles	# of Vehicles by Zone	# of Vehicles by Centroid	Total # of Vehicles	# of Vehicles by Zone	# of Vehicles by Centroid	Total # of Vehicles	# of Vehicles by Zone	# of Vehicles by Centroid	Total # of Vehicles	# of Vehicles by Zone	# of Vehicles by Centroid
Evacuation Zone #1	51736	2100	364	65	2100	266	55	2100	326	84	2100	384	83
	51731			95			73			95			97
	52830			204			139			147			204
Evacuation Zone #2	52833		843	53		911	44		881	76		837	74
	50686			48			71			108			116
	53129			221			153			144			164
	50680			57			63			55			57
	50689			43			52			45			42
	50677			39			49			43			39
	50674			54			61			57			58
	50671			53			62			57			58
	50668			47			54			47			44
	50662			73			94			84			78
	50665			43			57			49			44
	50659			54			67			53			31
	50656			56			83			63			33
Evacuation Zone #3	50634		893	64		923	61		893	52		879	54
	53136			367			237			191			216
	50640			48			64			86			100
	50637			56			68			64			65
	50643			52			64			85			101
	50647			79			121			144			136
	50650			93			132			149			141
	50653			135			176			122			65

Optimal vehicle routing, scheduling, and mobile charger dispatch during EV evacuations

presented by

Joseph Moyalan (University of California, Merced)
Xuchang Tang (University of California, Davis)
Qijian Gan (PATH/University of California, Berkeley)

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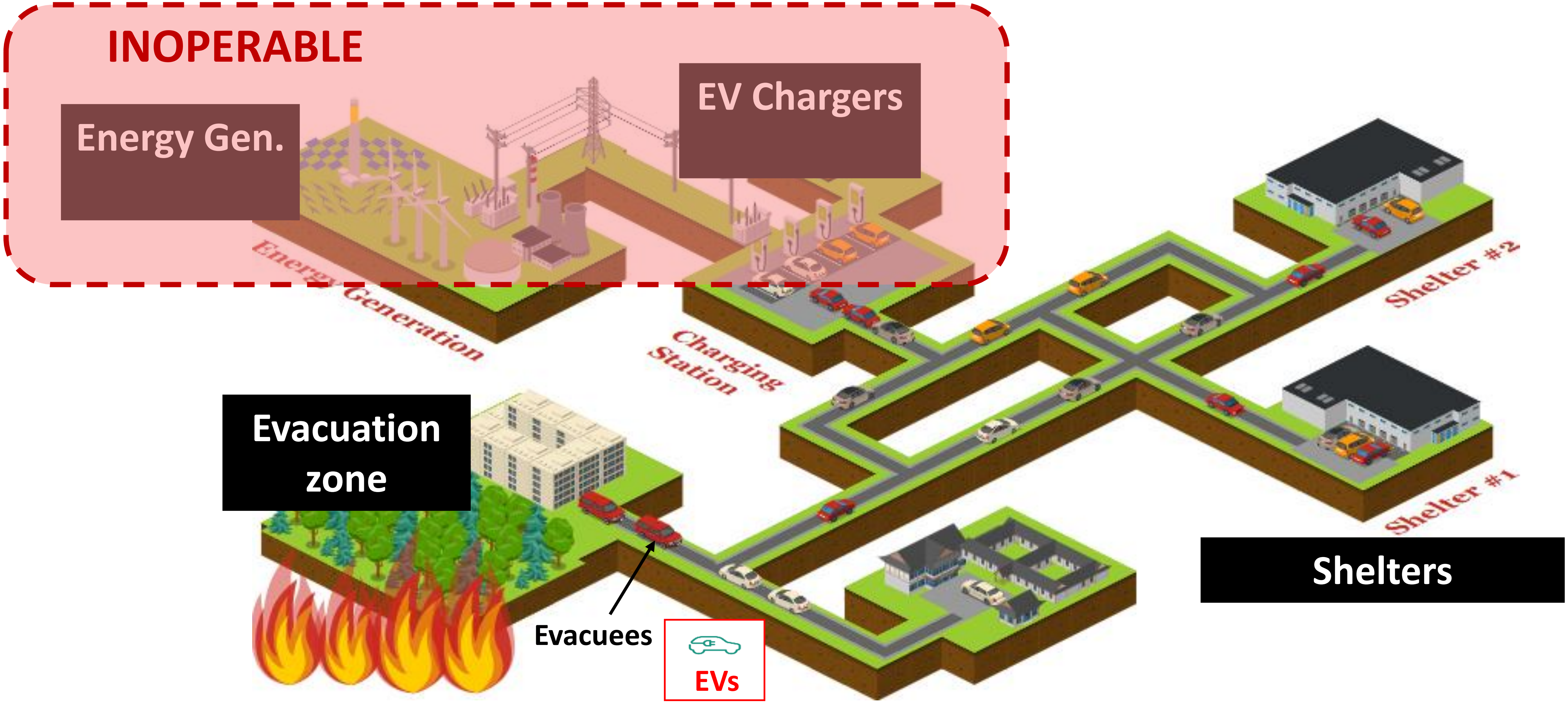
Part I

Optimal Planning of Emergency Evacuations Leveraging Equivalent Circuit Models

presented by

Joseph Moyalan
(University of California, Merced)

Evacuation: transportation, Charging & Energy



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Mobile Chargers Stations (MCS)

Several companies offer diverse set of MCS

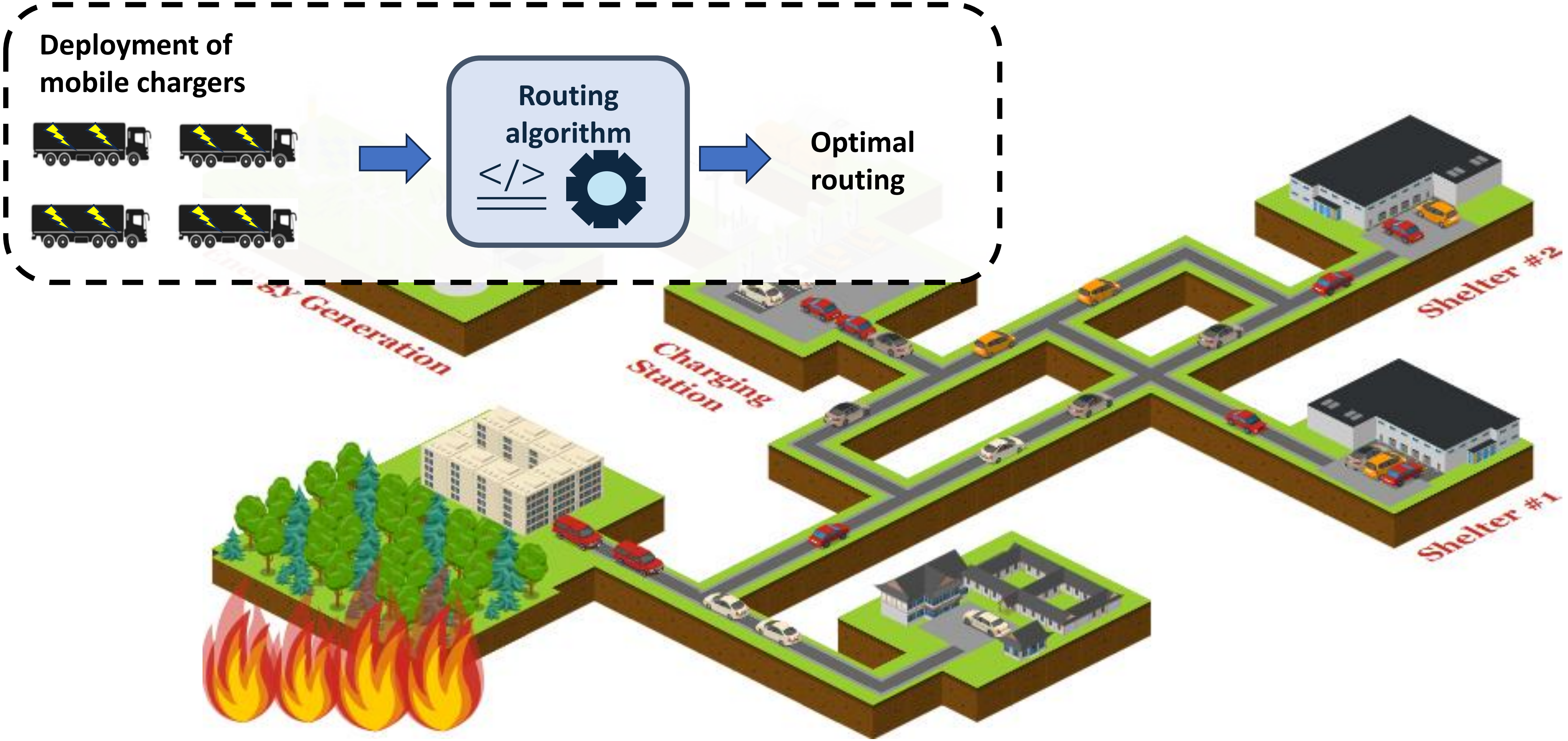
- Operate off-grid
- Can be easily re-deployed
- Level 2 and DCFC charging
- Energy storage:
  - Type: battery, H2, propane..
  - Size: From kWh to a few MWh

Source: https://moserenergy.com/products/rev-station/

Question: Can mobile chargers help during evacuation?

- Where to deploy?
- How many?
- What are the potential benefits?

Problem Formulation



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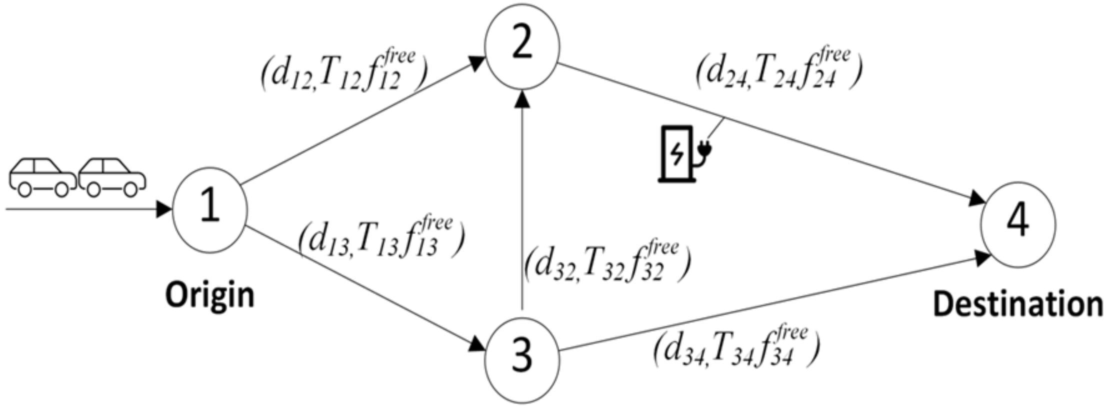
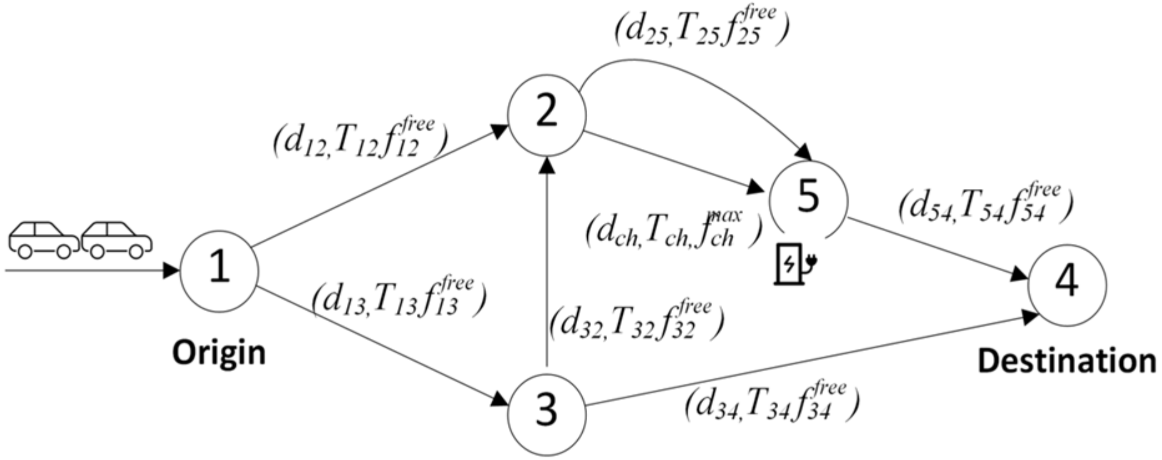
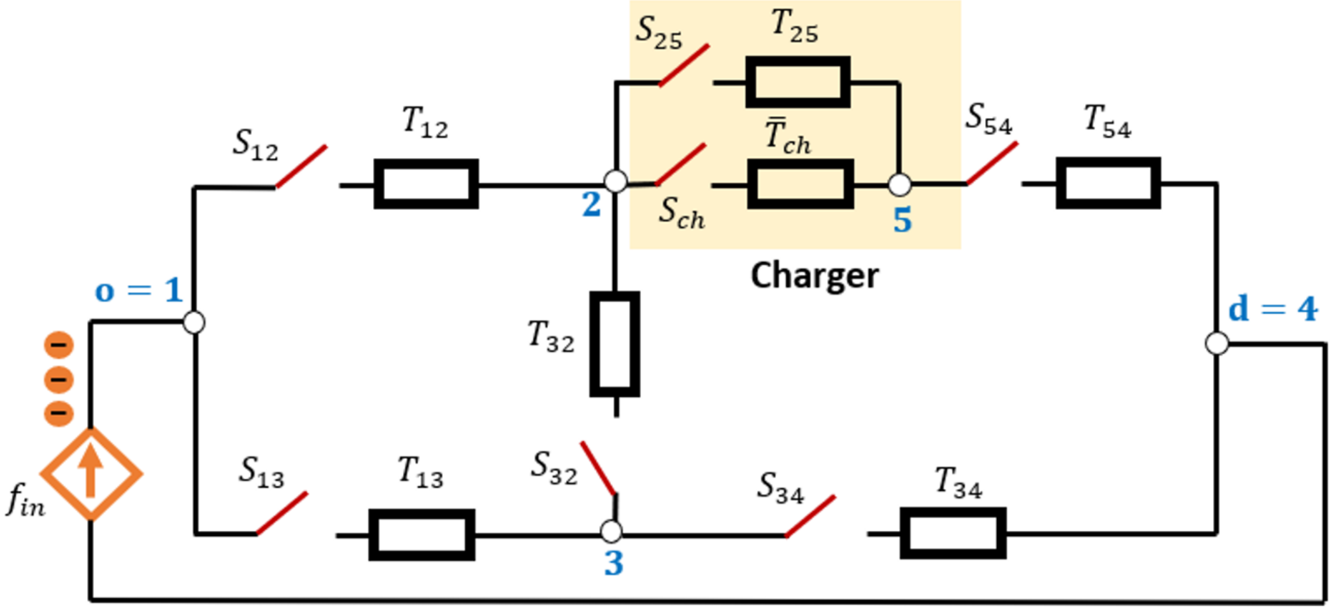
Equivalent Circuit Model (ECM) for Transportation Network







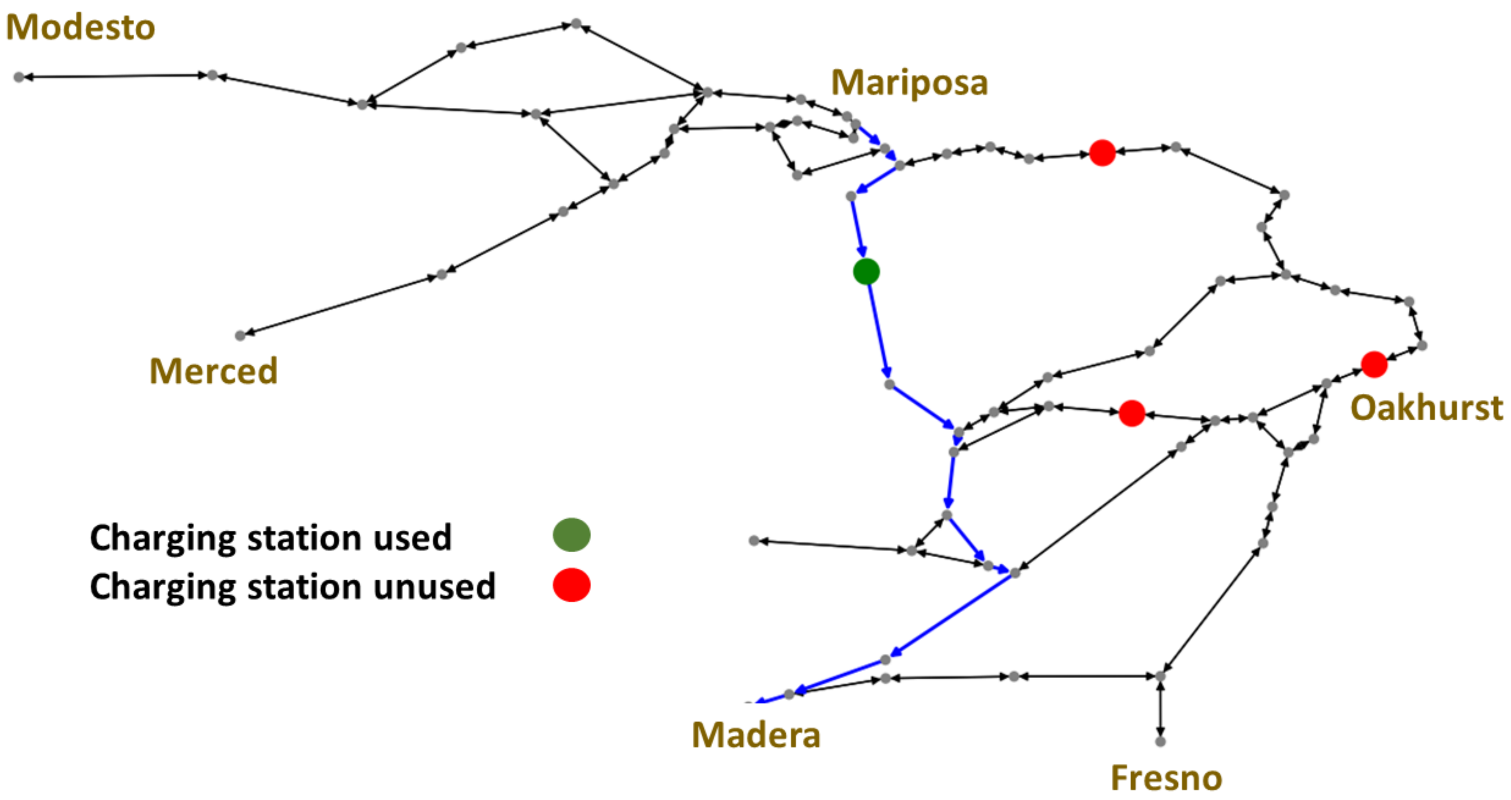
ECM model

Decision variables are S_{ij} and S_{ch} which take Boolean values $\{0,1\}$

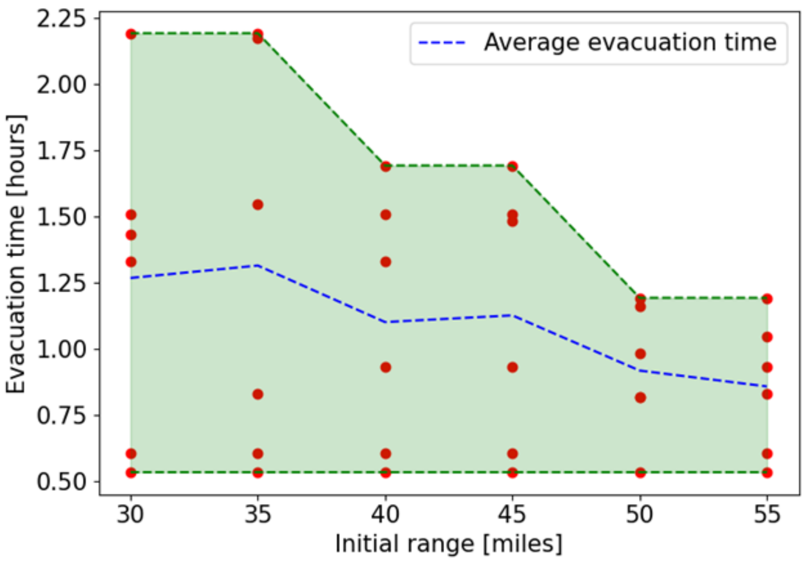
Mariposa network

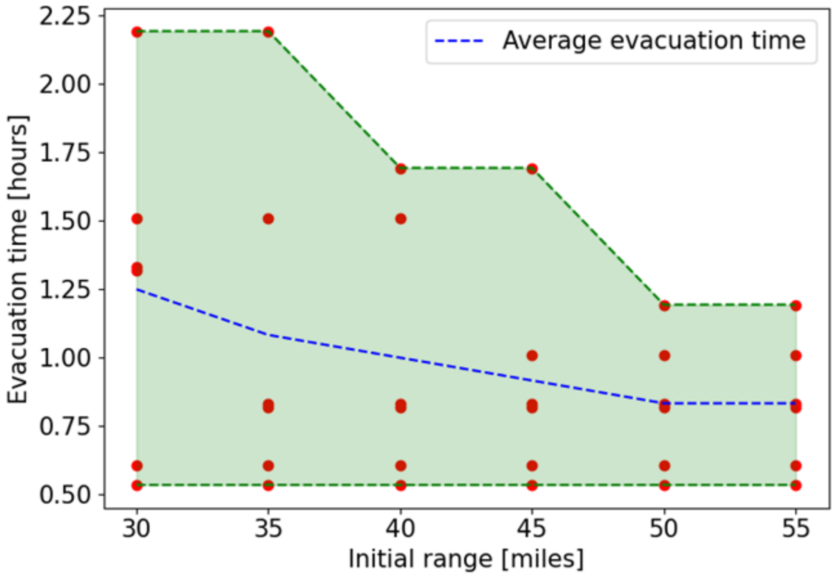
Parameters

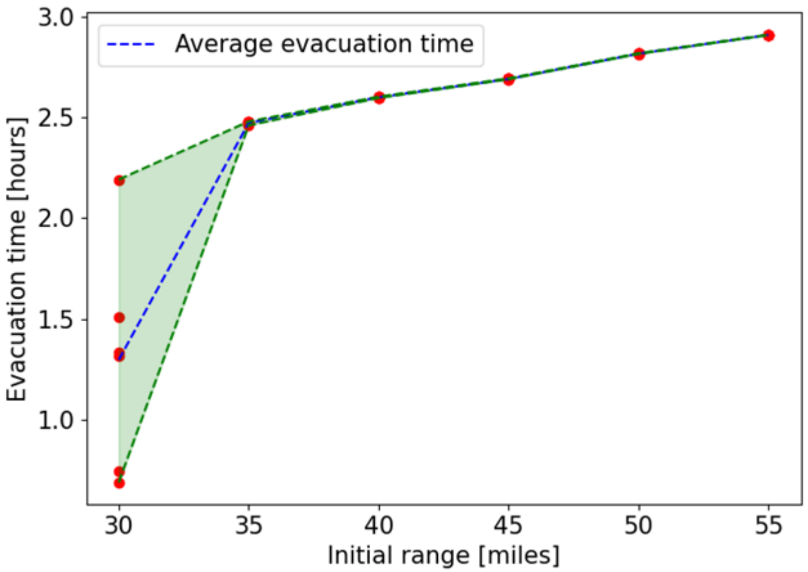
- Number of nodes: 61
- Number of links: 71
- Number of (od)-pairs: 6
- Charging speed: 150 kWh
- f_{ij}^{free}: 1000 vehicles/hr
- f_{in}: 50 vehicles/hr



Sensitivity analysis





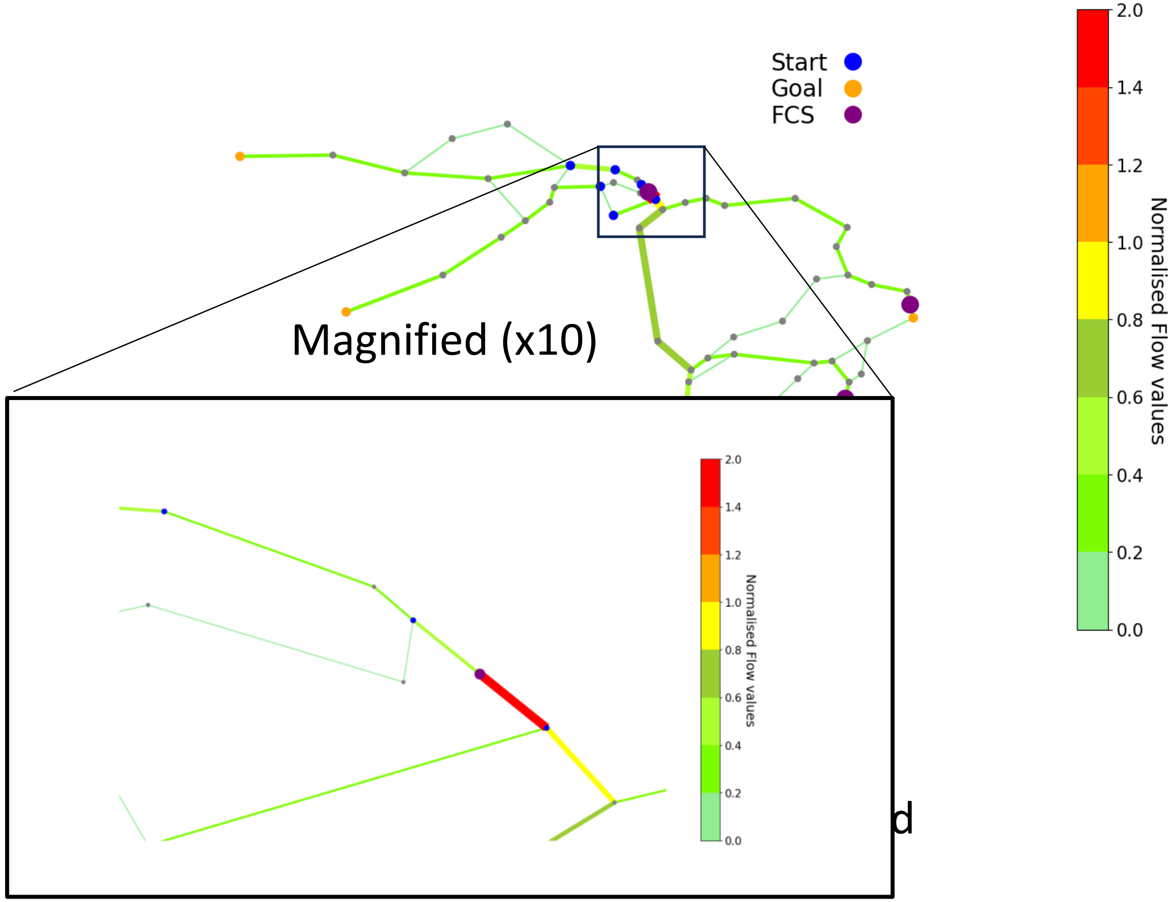


$$J^{max} = t_{evac,m}(\mathbf{S}_m, \mathbf{S}_m^{ch})$$

$$J^{avg} = \frac{1}{M}\sum_{m} t_{evac,m}$$

$$J^{\Delta} = \max_{m}\left|t_{evac,m} - J^{avg}\right|$$

Comparison with Uncoordinated Evacuation



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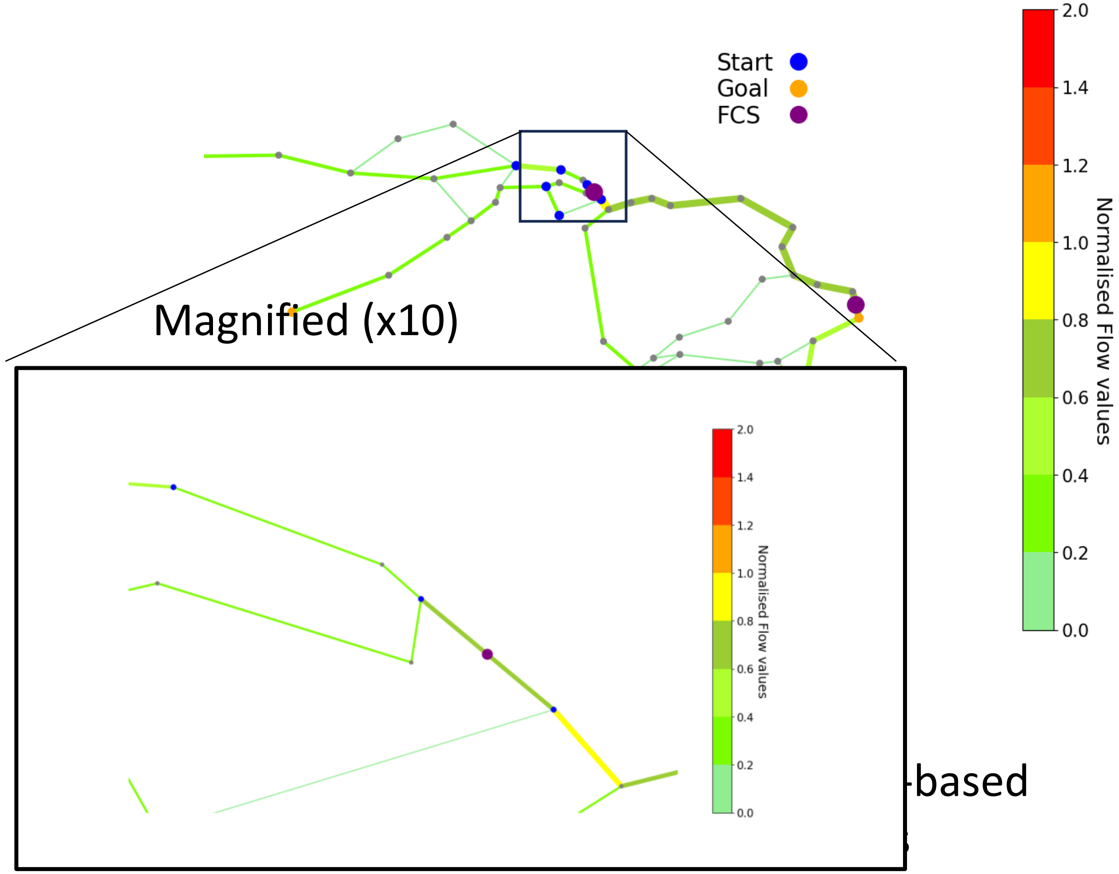
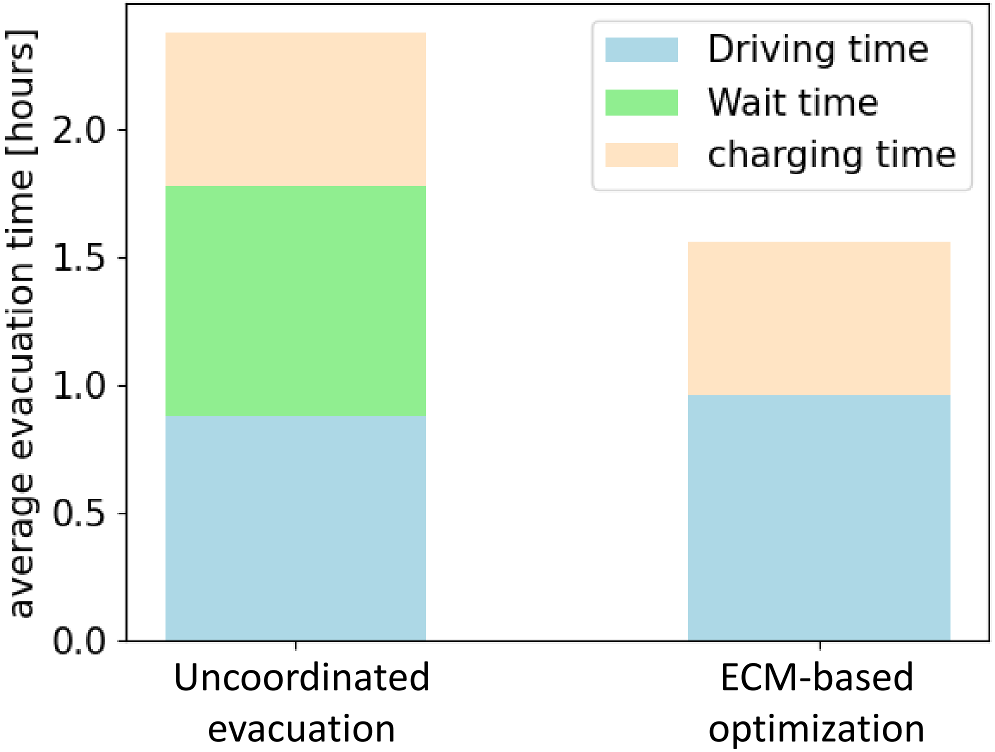
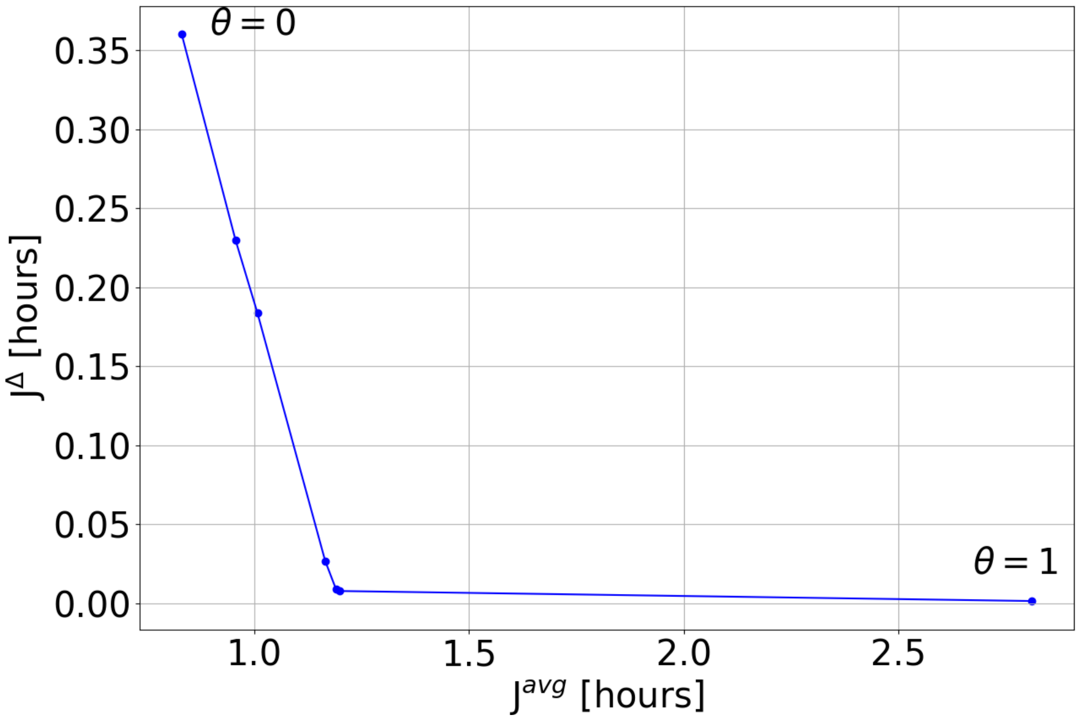


-based

Sensitivity analysis



Average evacuation time comparison



Values of the performance indexes for different range of θ ($r_0 = 50$ miles) obtained with $\mathbb{P}^{avg+\Delta}$

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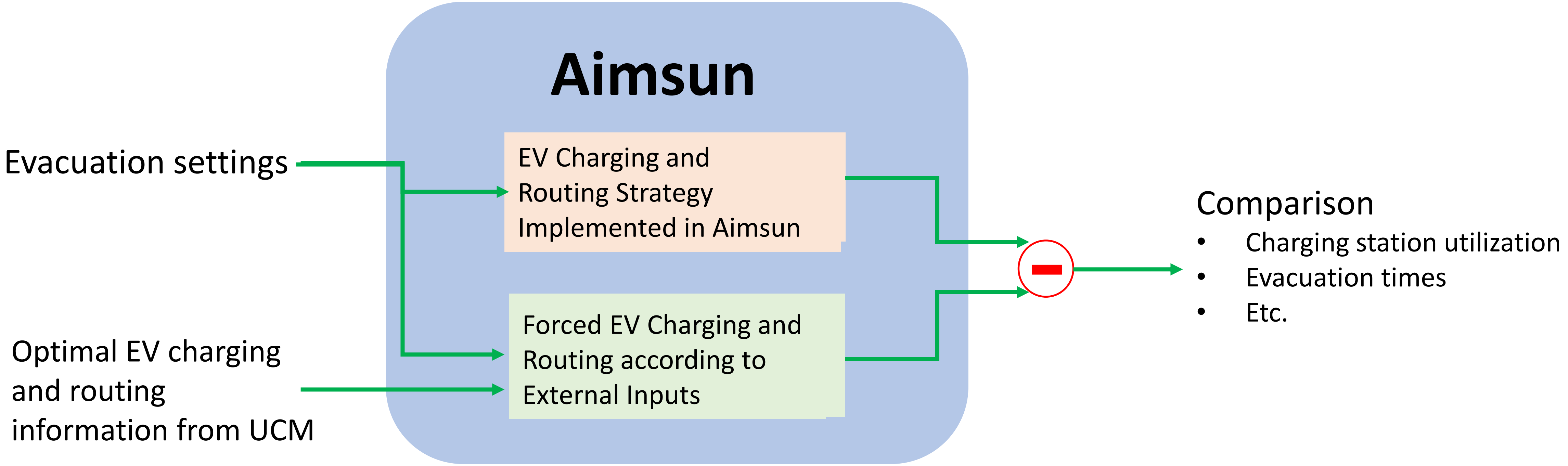
Part II

Visualization and Validation of Optimization Results in Aimsun

Presented by

Qijian Gan (PATH/University of California, Berkeley)

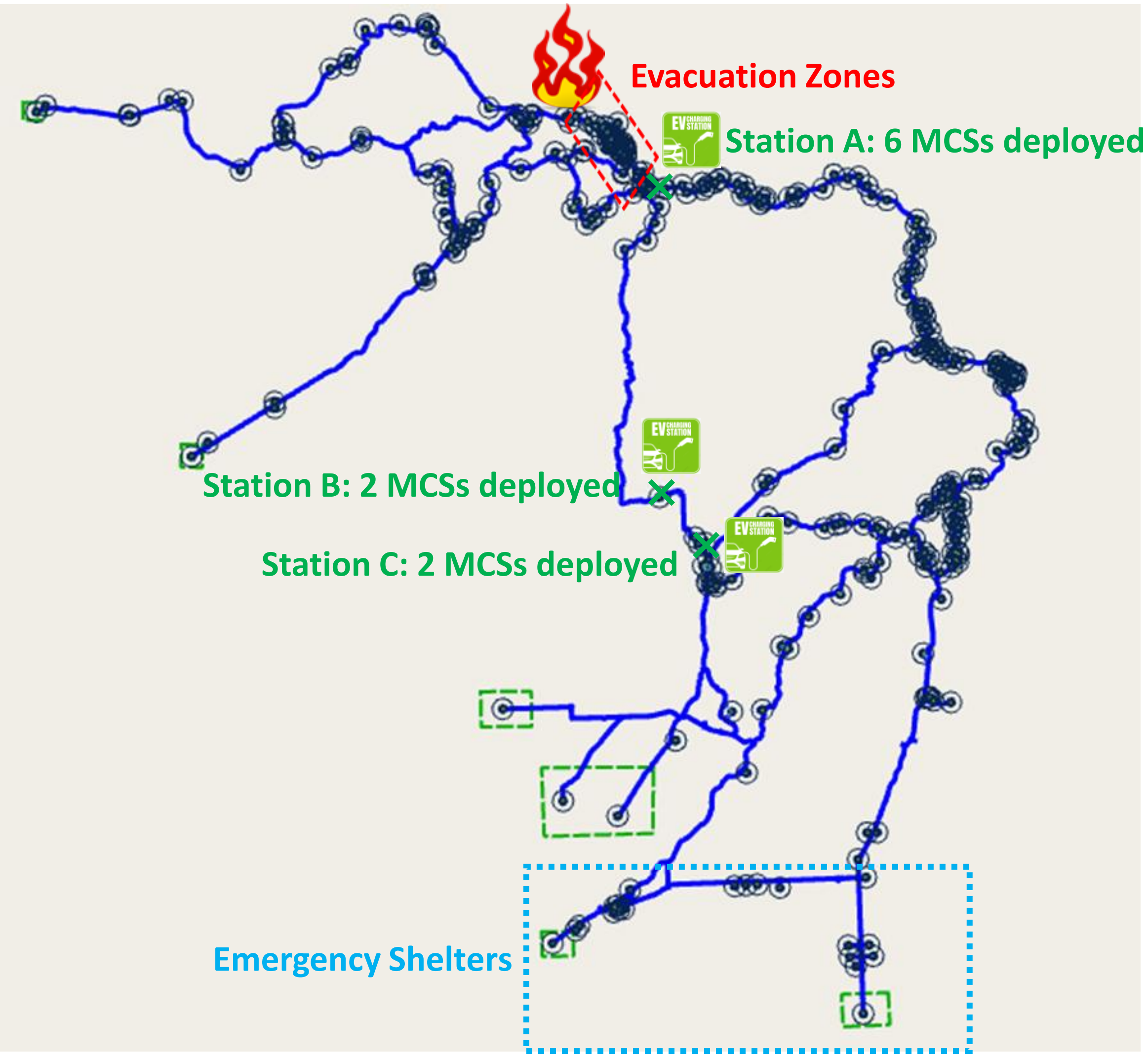
Validation Framework



EV Charging and Routing Strategy Implemented in Aimsun

- The estimated vehicle queueing time is considered at each charging station.
- An EV will pick a charging station with the earliest service time.

Implementing UCM's Optimization Results



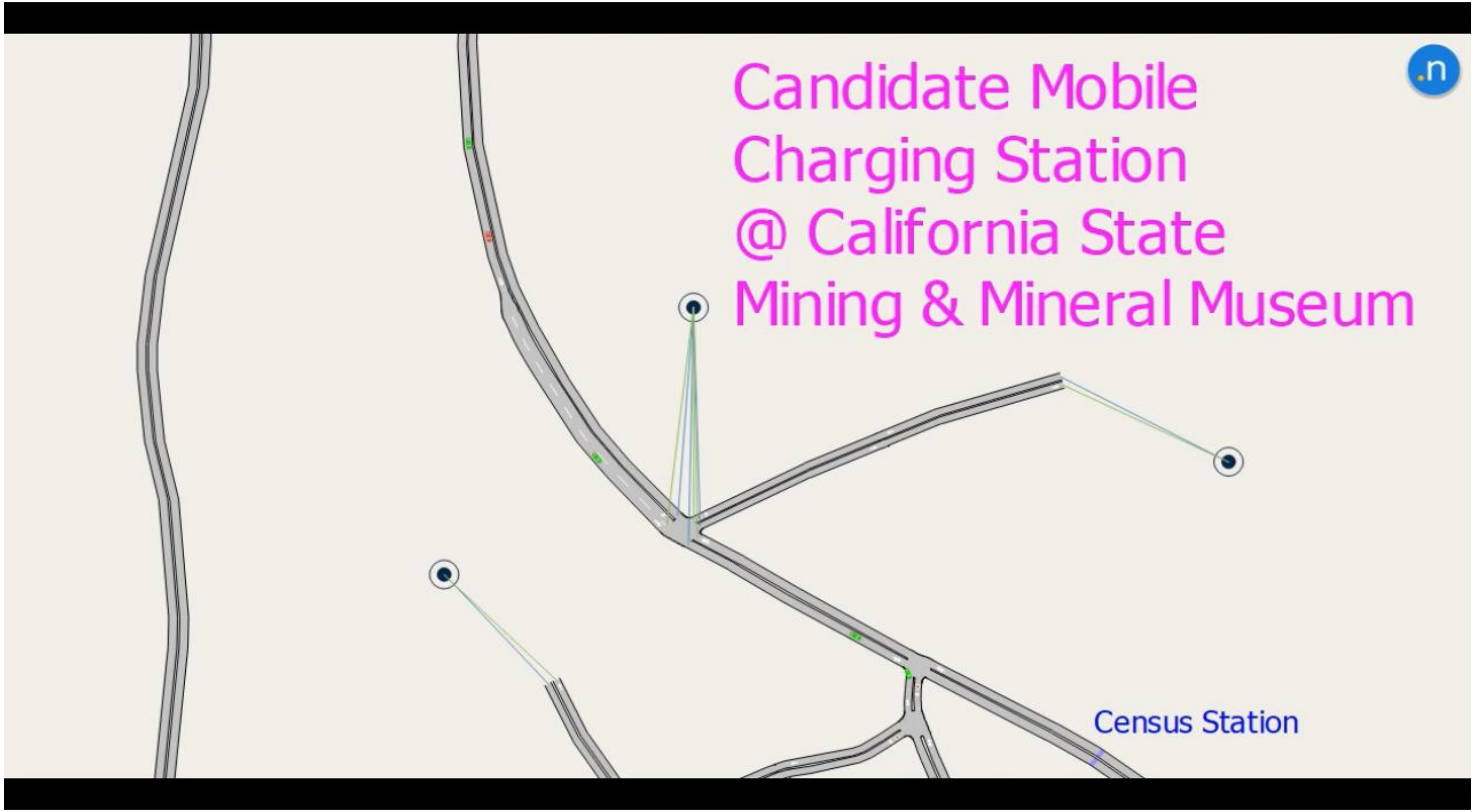
Evacuation Demand:

- 2000 vehicles to be evacuated in a 6-hour period
    - 500 vehicles per hour to be evacuated in the first 4 hours
- 100% EVs
    - 200 EVs (i.e., 10% of the total EVs) with a low SOC of 20%
    - The rest with a high SOC greater than 50%

Recommended Deployment of MCSs

- Three candidate MCS locations along the evacuation route
    - Station A: 6 MCSs with 5 ports and 750 kWh battery capacity
    - Station B: 2 MCSs with 5 ports and 750 kWh battery capacity
    - Station C: 2 MCSs with 5 ports and 750 kWh battery capacity

Example: EV Charging Patterns at Station A



Legend
- Red: SOC <= 20%
- Yellow: 20%<SOC<=60%
- Green: SOC > 60%

- An EV with an SOC less than 20% will enter the charging station, disappearing from the intersection.
- An EV with a targeted SOC (e.g., 50% in this example) will be released from the charging station, entering one of the sections connected to the intersection.

Example: Charging Station Status at Station A

```
Simulation Time(s)= 7800.00
Actual Time(in 24hr)= 8:10
Charging Station Id= 51193
Mobile Charging Station Ids= 5119301;5119302;5119303;5119304;5119305;5119306;
DCFC Fixed {Station Id:Charger Id}=
L2 Fixed {Station Id:Charger Id}=
DCFC Mobile {Station Id:Charger Id}= 5119301:1;5119301:2;5119301:3;5119301:4;5119301:5;5119302:6;5119302:7;5119302:8;5119302
L2 Mobile {Station Id:Charger Id}=
Total battery consumed (kwh; fixed charging station)= 0
Total battery consumed (kwh; mobile charging stations)= 1740.28
Remaining battery {mobile station id:remaining battery(kwh)}= 5119301:418.415;5119302:418.712;5119303:419.729;5119304:451.35
Available DCFC Chargers= 19;20;21;22;23;24;25;26;27;28;29;30;1;2;3;4;5;6;7;8;9;10;11;
Available L2 Chargers=
Inactive DCFC Chargers=
Inactive L2 Chargers=
{DCFC Charger ID--EV ID--Expected Available Time(s)}= 12-1015-7888.06;13-987-7902.97;14-989-7931.05;15-997-8004.94;16-1030-8
{L2 Charger ID--EV ID--Expected Available Time(s)}=
DCFC Served Vehicles= 6;19;45;62;30;40;43;46;55;74;83;86;89;112;139;130;128;182;200;189;214;229;262;253;261;272;267;290;293;
L2 Served Vehicles=
DCFC Waiting Vehicles=
L2 Waiting Vehicles=
DCFC Reserved Vehicles (optional)= 6;19;45;30;62;43;40;46;55;74;83;86;89;112;139;130;128;182;200;189;214;229;262;253;261;272
L2 Reserved Vehicles (Optional)=
```

Time

Station ID

MCS IDs

Battery Consumption

MCS Remaining Battery

Available Chargers

Chargers Being Used

EVs Released to the Network after Charging

EVs Routed to the Charging Station

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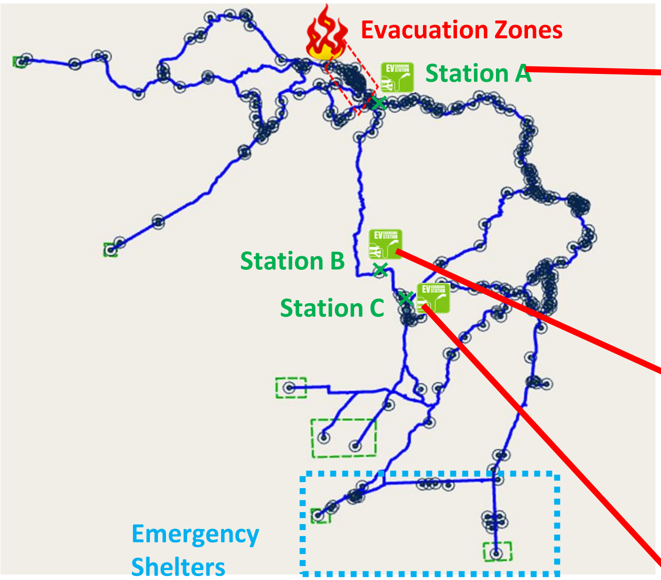
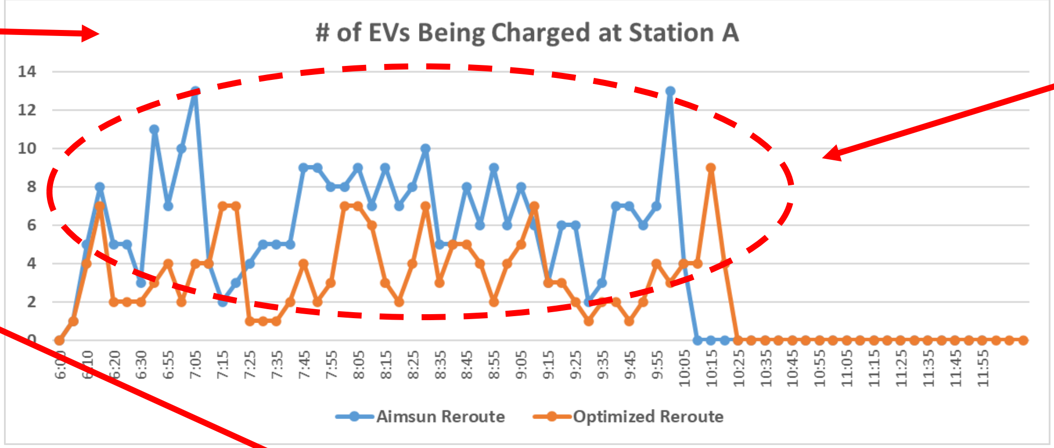
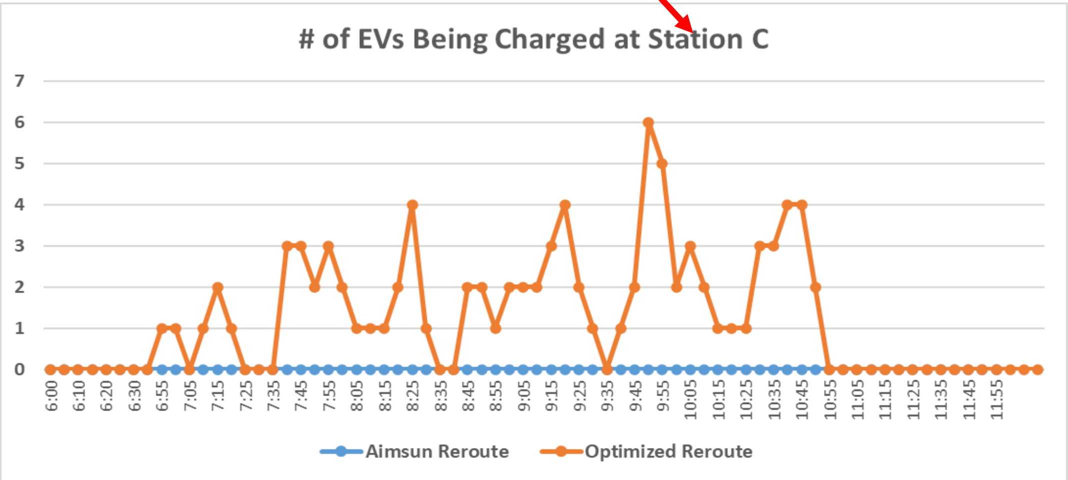
Comparison of Charging Station Management

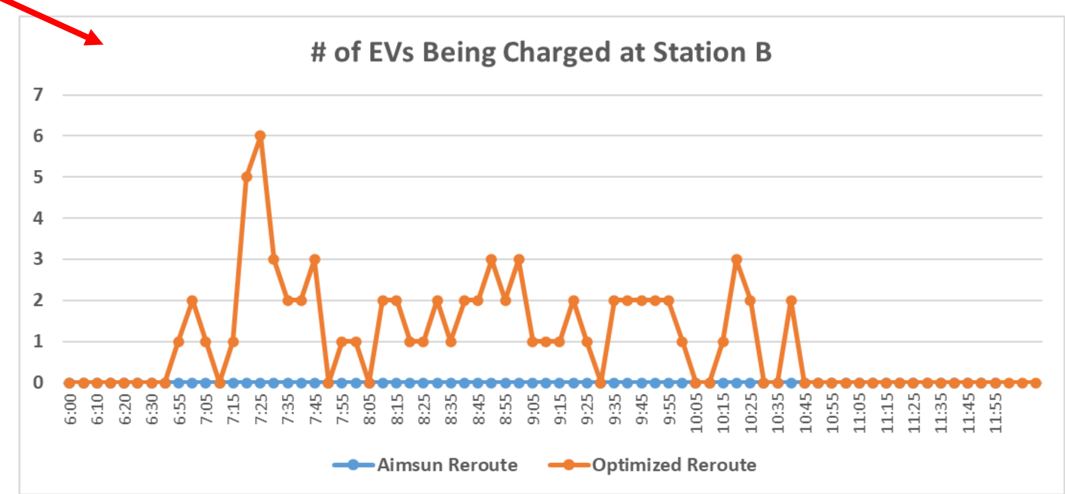




Observations:

- Early charging is important to ensure EVs to have enough battery to reach the destination or the next charging station.
- The algorithm implemented in Aimsun tends to route EVs to the nearest charging stations with the earliest time to be served.
- The optimization algorithm proposed by UCM tends to balance the charging demand along the detour route to avoid potential queueing impact at certain charging stations.





Comparison of Evacuation Times

Origin	Destination	Initial SOC (%)	Averaged Evacuation Time (secs)					
			Optimization Algorithm		Optimized EV Routing in Aimsun		Implemented EV Routing in Aimsun	
Evacuation Zone #1 (51736)	To Madera (50623)	20	Charging Station A	4890	Charging Station A	4462	Charging Station A	4467
Evacuation Zone #1 (51731)	To Rolling Hills (50626)	20	Charging Station B	5438	Charging Station B	5130	Charging Station A	4896
Evacuation Zone #2 (50668)	To Madera (50623)	>50		2876		3910		3922
Evacuation Zone #2 (50662)	To Rolling Hills (50626)	>50		3414		4287		4297
Evacuation Zone #2 (50659)	To Madera (50623)	20	Charging Station C	4601	Charging Station C	4523	Charging Station A	4207
Evacuation Zone #2 (50677)	To Rolling Hills (50626)	>50		3451		4335		4346
Evacuation Zone #3 (50637)	To Madera (50623)	>50		2912		3927		3931
Evacuation Zone #3 (50640)	To Rolling Hills (50626)	>50		3475		4351		4373
Evacuation Zone #3 (50647)	To Madera (50623)	20	Charging Station A	4652	Charging Station A	4273	Charging Station A	4274
Evacuation Zone #3 (50653)	To Rolling Hills (50626)	>50		3382		4250		4271
Averaged Evacuation Time (secs)					4224 (1990 vehs)		4223 (1969 vehs)	

Because there are plenty of MCSs available for early charging, we do not see significant differences between the two routing algorithms.

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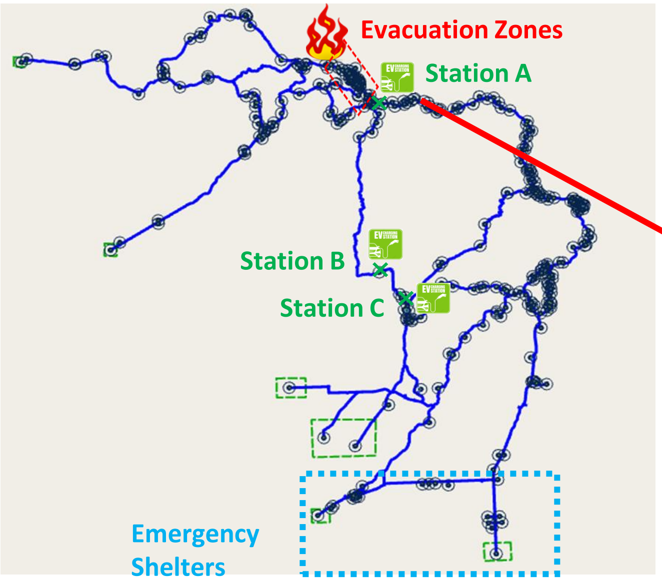
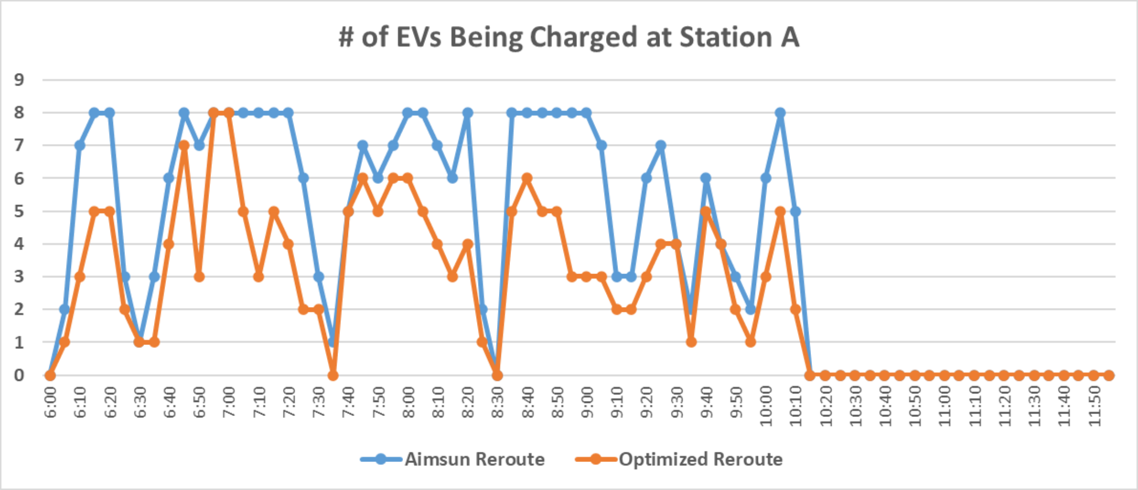
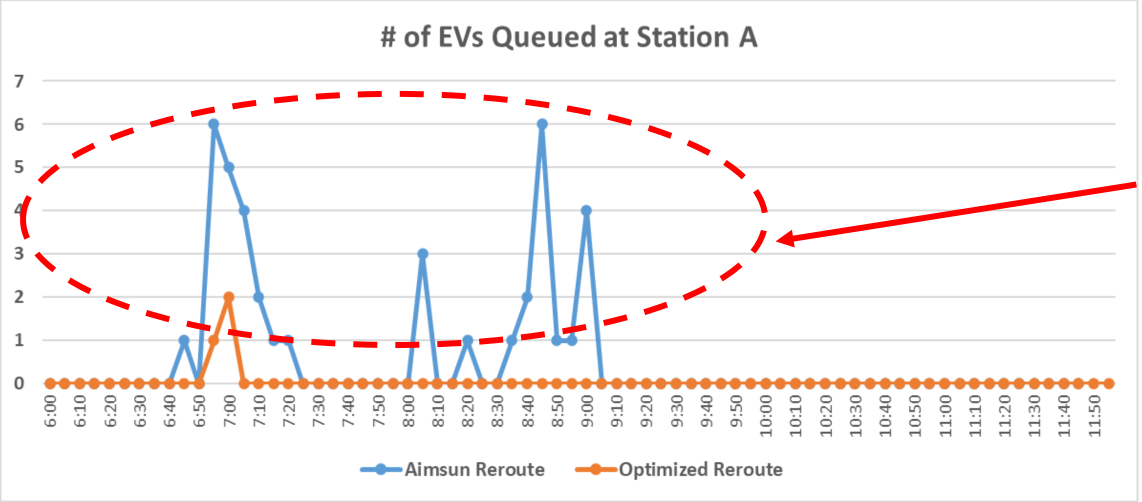
When Less MCSs are Available for Early Charging (1/2)





Station A:
6 MCSs with 5 ports --> 2 MCSs with 4 ports



EVs are queueing up at Station A when they are rerouted by the algorithm implemented in Aimsun.

When Less MCSs are Available for Early Charging (2/2)

Origin	Destination	Initial SOC (%)	Averaged Evacuation Time (secs)					
			Optimization Algorithm		Optimized EV Routing in Aimsun		Implemented EV Routing in Aimsun	
Evacuation Zone #1 (51736)	To Madera (50623)	20	Charging Station A	4890	Charging Station A	4433	Charging Station A	4510
Evacuation Zone #1 (51731)	To Rolling Hills (50626)	20	Charging Station B	5438	Charging Station B	5060	Charging Station A	5000
Evacuation Zone #2 (50668)	To Madera (50623)	>50		2876		3872		3881
Evacuation Zone #2 (50662)	To Rolling Hills (50626)	>50		3414		4259		4274
Evacuation Zone #2 (50659)	To Madera (50623)	20	Charging Station C	4601	Charging Station C	4299	Charging Station A	4303
Evacuation Zone #2 (50677)	To Rolling Hills (50626)	>50		3451		4335		4362
Evacuation Zone #3 (50637)	To Madera (50623)	>50		2912		3899		3912
Evacuation Zone #3 (50640)	To Rolling Hills (50626)	>50		3475		4373		4404
Evacuation Zone #3 (50647)	To Madera (50623)	20	Charging Station A	4652	Charging Station A	4264	Charging Station A	4341
Evacuation Zone #3 (50653)	To Rolling Hills (50626)	>50		3382		4236		4255
Averaged Evacuation Time (secs)					4195		4213	

In this case, shifting the charging demand to downstream charging stations, for example, using the algorithm proposed by UCM, becomes more appropriate.

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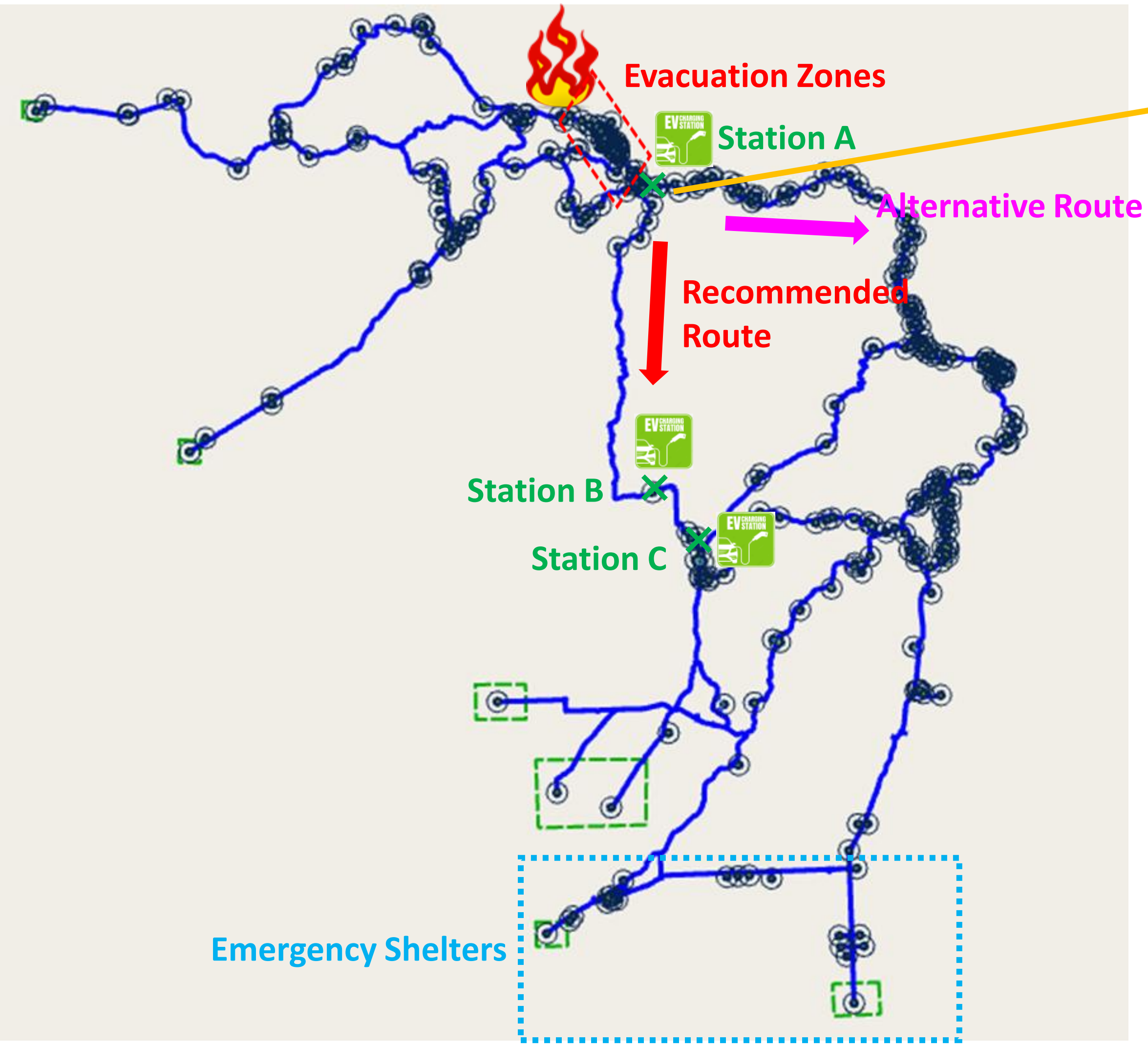
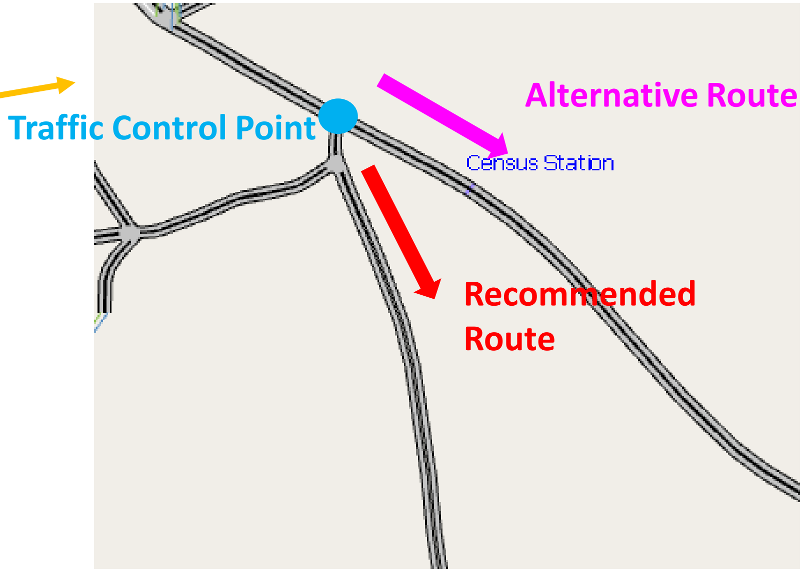
Practical Traffic Management Considerations





- We find that both UCM's optimization algorithm and Aimsun tend to recommend EVs to use a shorter route, which has less infrastructure support.
- This may congest the roads further downstream and induce traffic accidents. In this case, it will make traffic evacuation become more difficult.
- Therefore, we need to apply traffic control at key locations to detour a certain percentage of the traffic to the alternative routes.

Part III

Vehicle-Based MCS-Included Optimal Evacuation Planning using Linear Programming

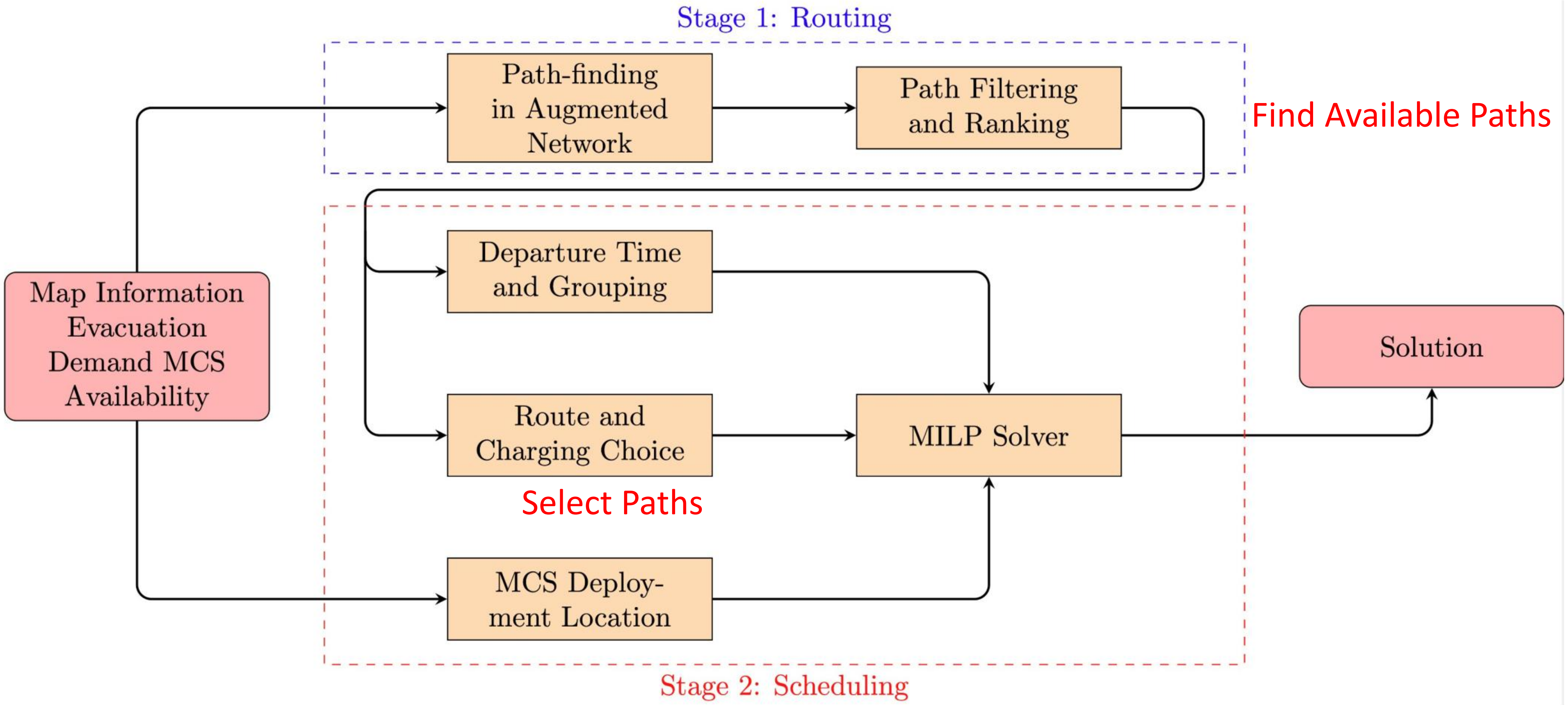
Presented by

Xuchang Tang
University of California, Davis

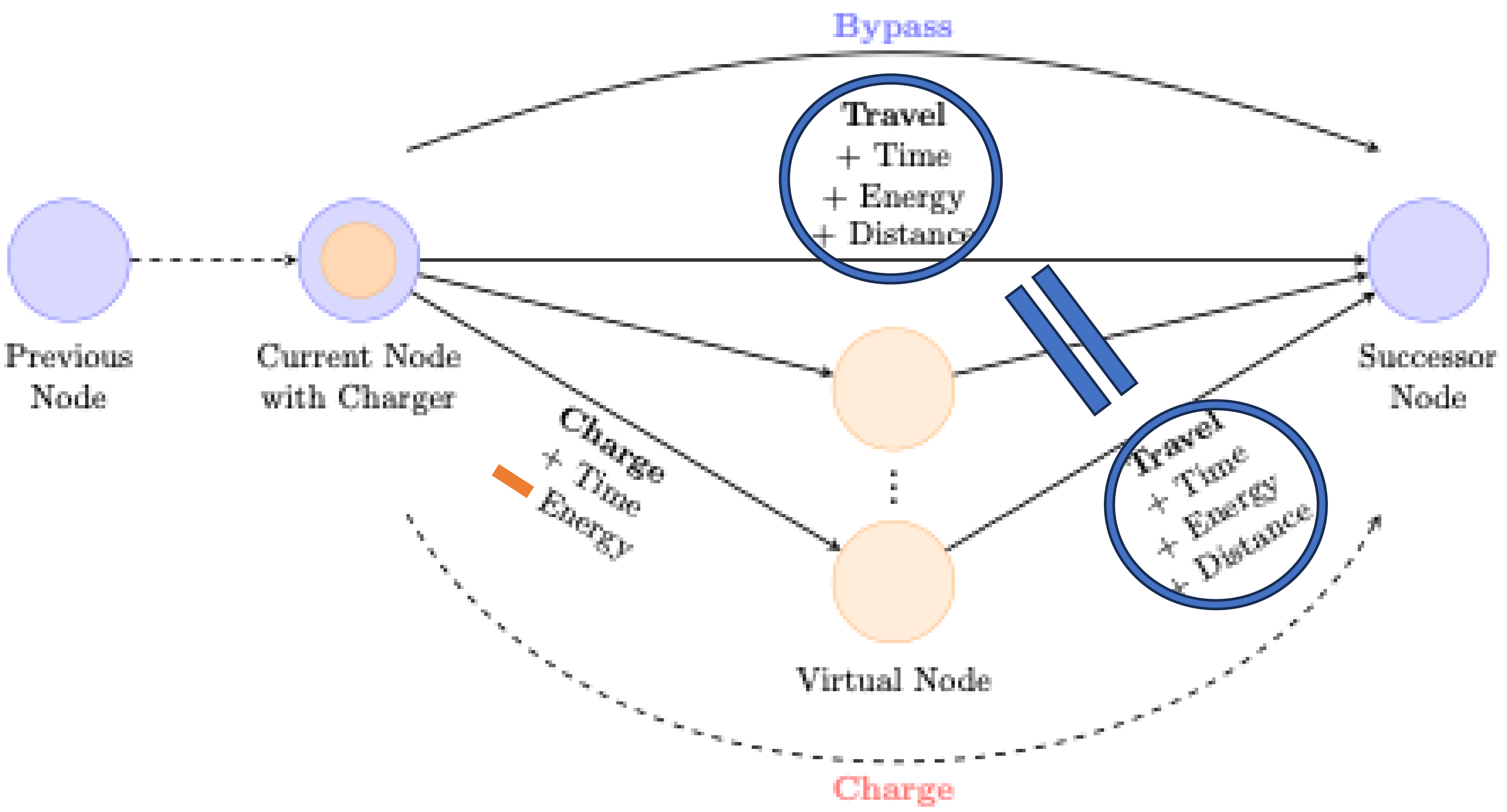
Two-Staged Approach

Routing Charging

Evacuation Path: traveling **waypoints** + **charging** options



Network Augmentation to Represent Charging



Mixed Integer Linear Programming Formulation

Objective: Minimize **Average** Evacuation Time

$$\min \sum_{w \in \mathbb{W}} \sum_{p \in \hat{\mathbb{P}}_w} \sum_{t \in \mathbb{T}} (t + \tilde{t}_{wpt}) \cdot x_{wpt}$$

Constraint #1: Satisfy evacuation demand

$$\sum_{p \in \hat{\mathbb{P}}_w} \sum_{t \in \mathbb{T}} x_{wpt} = f_w, \quad \forall w \in \mathbb{W}$$

Constraint #2: Physical deployment of MCS

$$\sum_{n \in \mathbb{N}} q_{mn} = 1, \quad \forall m \in \mathbb{M}$$

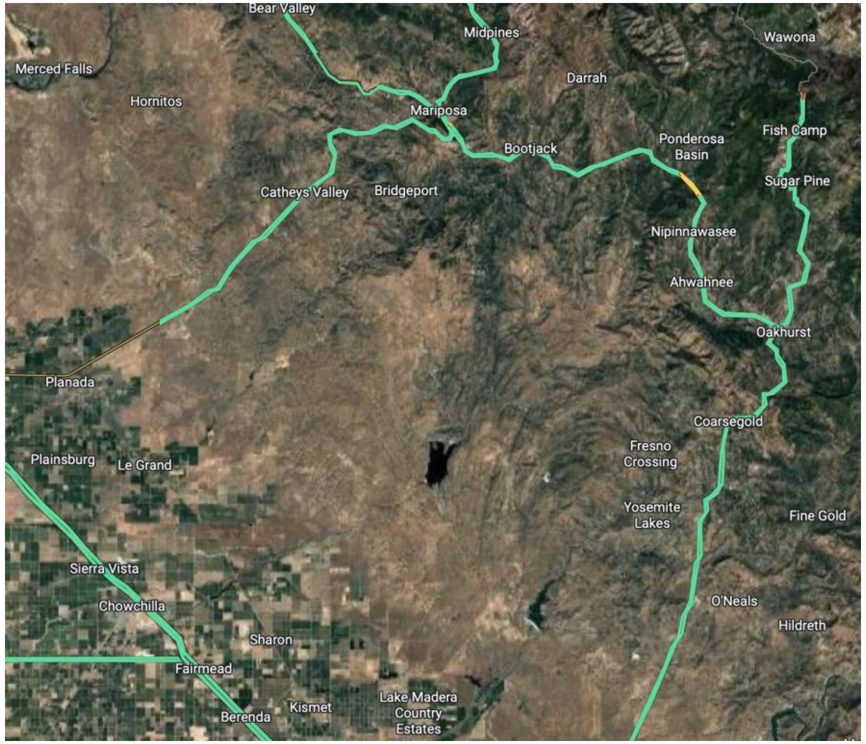
Constraint #3: The number of charging port supplied upper-bounds the charging demand

$$D_{nt} \leq S_{nt}, \quad \forall n \in \mathbb{N},\ \forall t \in \mathbb{T}$$

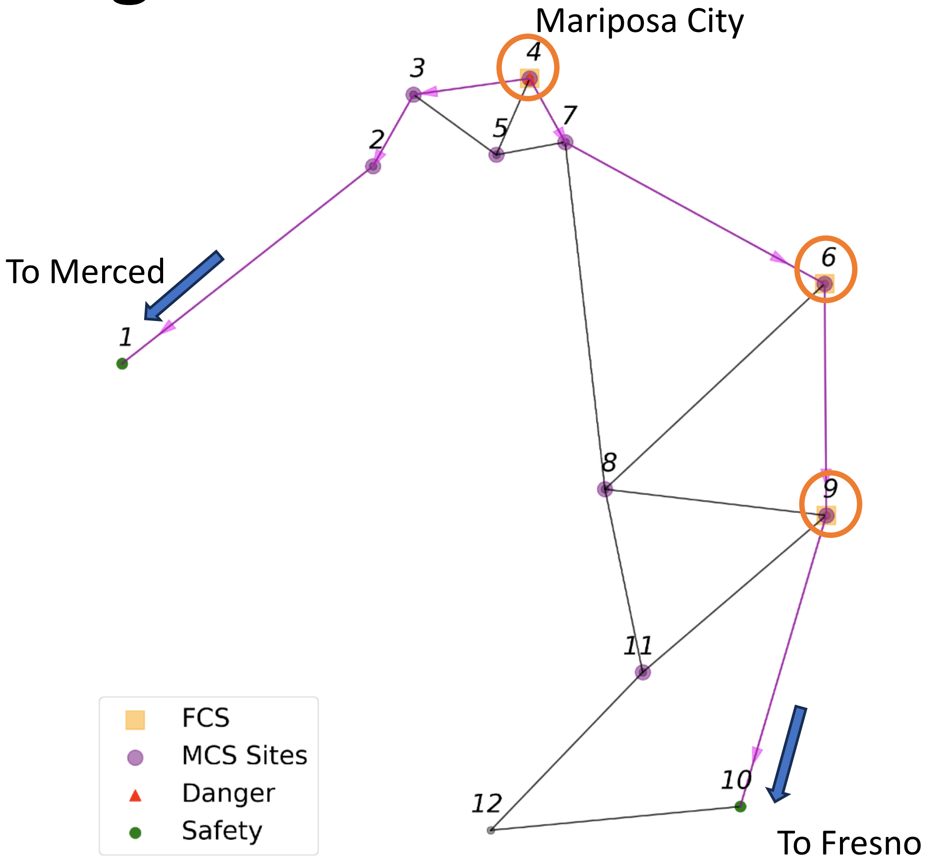
Constraint #4: MCS energy is bounded between 0% and 100%

$$0 \leq soc_{mt} \leq 1, \quad \forall m \in \mathbb{M},\ \forall t \in \mathbb{T}$$

Case Study: Mariposa Region



Satellite image of the Mariposa region



Abstracted Network used in Optimization

Case	**Battery Size (kWh)**	**Initial SOC**
Homogeneous	30 (Fixed)	0.2 (Fixed)
Heterogeneous	Uniform $\mathcal{U}\{20, 30, 40\}$	Truncated Normal $\mathcal{N}_T(0.2, 0.1, [0.0, 1.0])$

Parameters for **Heterogeneous** Demand

Evacuation Demand

From ***Mariposa City***

300 towards ***Merced***

300 towards ***Fresno***

(Homogeneous)

30 kWh Battery/vehicle

20% initial SOC

20 kWh per 100 km

Charging Infrastructure

Node #4: 20 FCS Port

Node #6: 20 FCS Port

Node #9: 40 FCS Port

Rate: 48 kW

MCS

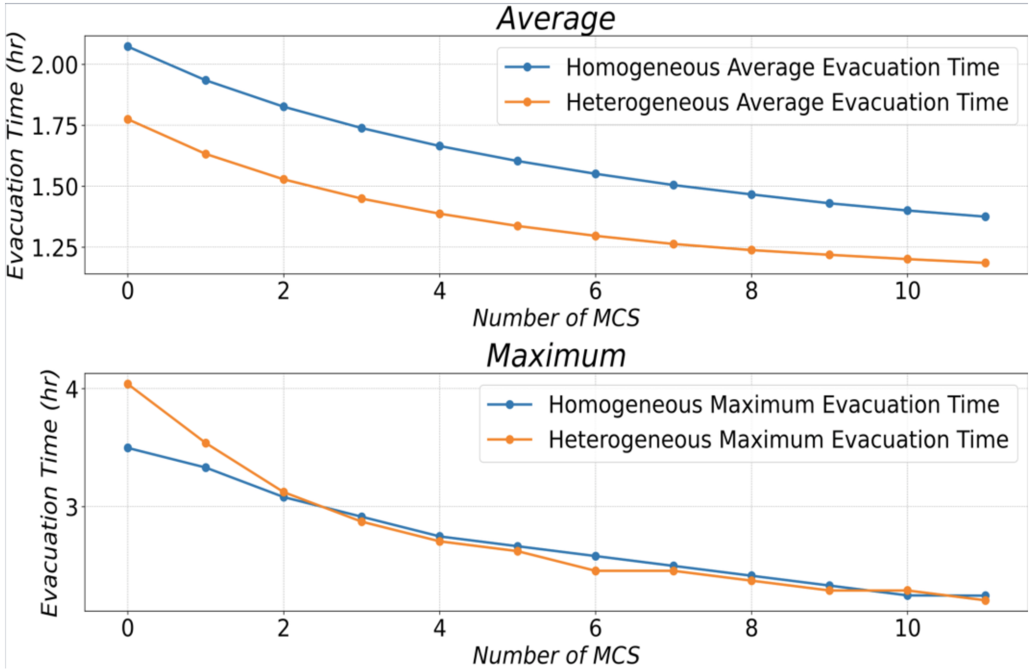
Ports: 5 per MCS

Capacity: 420 kWh

Rate: 48 kW

Site limit: 5 MCS max

MCS significantly accelerates evacuation operation



Evacuation time vs. No. of MCS

Evacuation time: Start of evacuation → Arrival

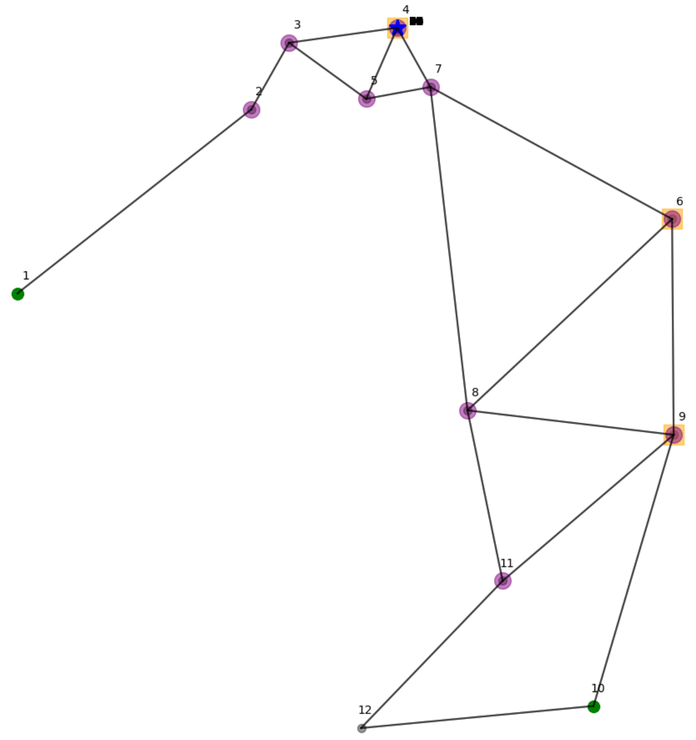
- Wait + charging + travel;
- 2 MCS (+12.5% in port) -> -25% in max time (hetero)
- 4 MCS (+25%): -33% in max time (hetero)

Wait time: Start of evacuation → Departure of the vehicle

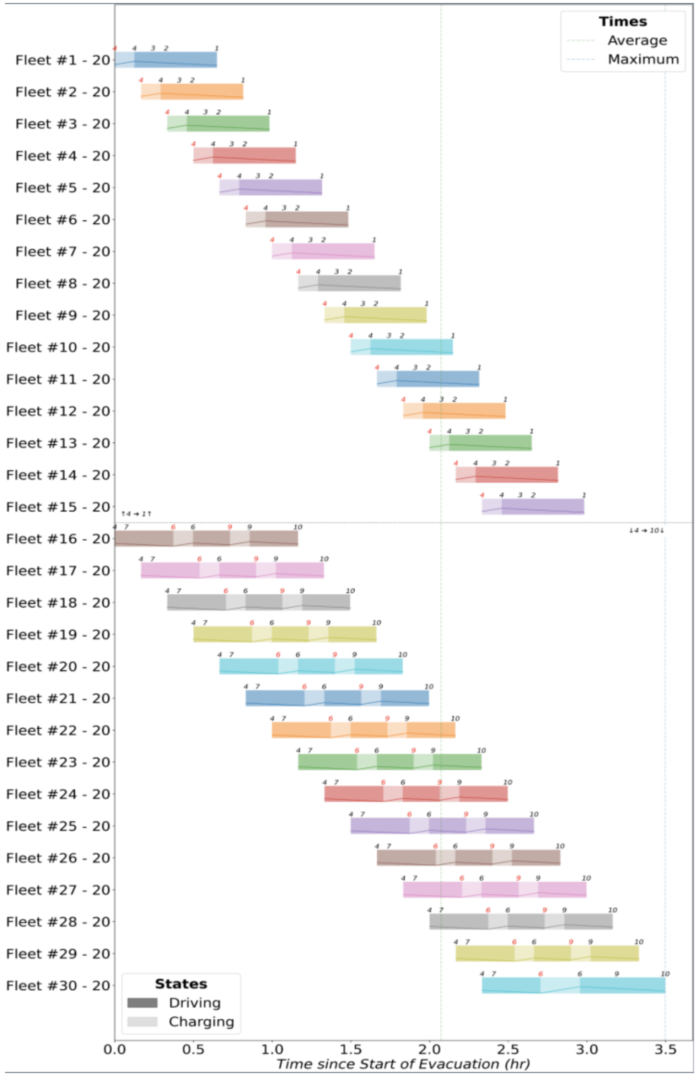
- Main time component that MCS can cut;
- 1 MCS (+6.25% in port) -> -15% on average;
- 4 MCS (+25%): -35% (uniform), -44% (hetero) on average

MCS could enable disproportionate reduction in evacuation time! Why and how?

Patterns of Optimal Routing and Scheduling



Animation of the Evacuation Progress with 0 MCS



Example of Evacuation Progress with 0 MCS

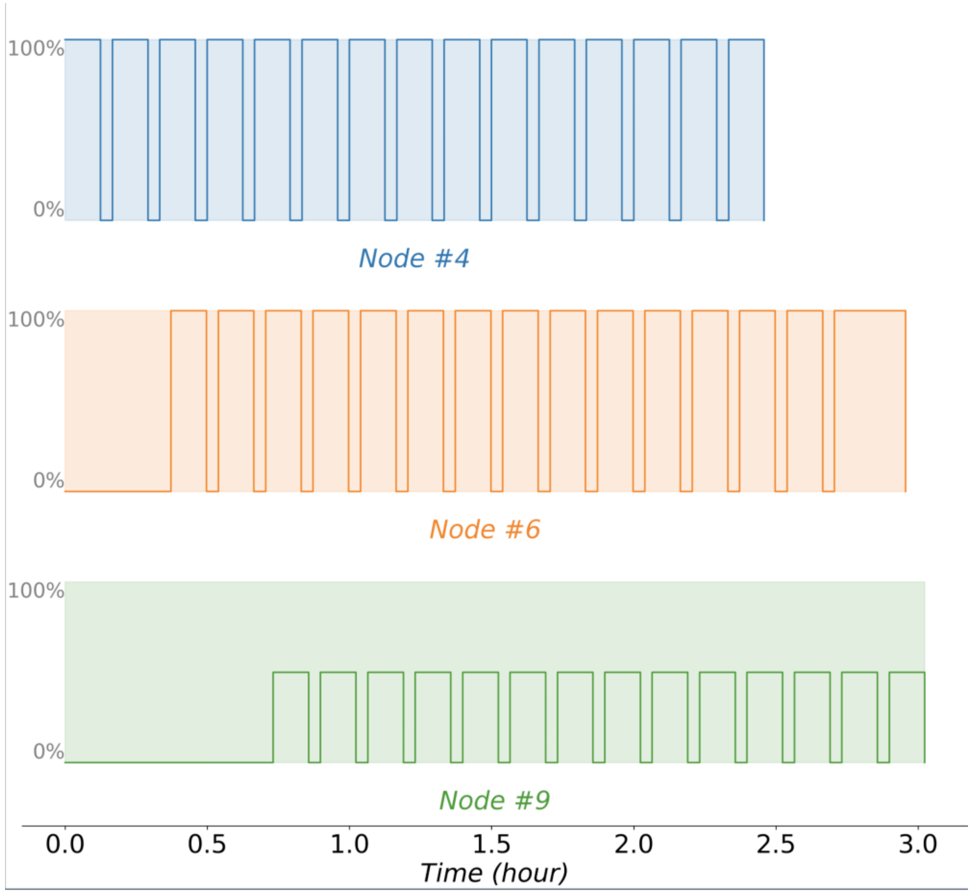
Features of optimal evacuation schedule:

- Gantt Chart: schedule of departure, travel, and charging;
- “Staggered” pattern: fleets arriving at charging stations in order;
- Charging facility is the limiting factor

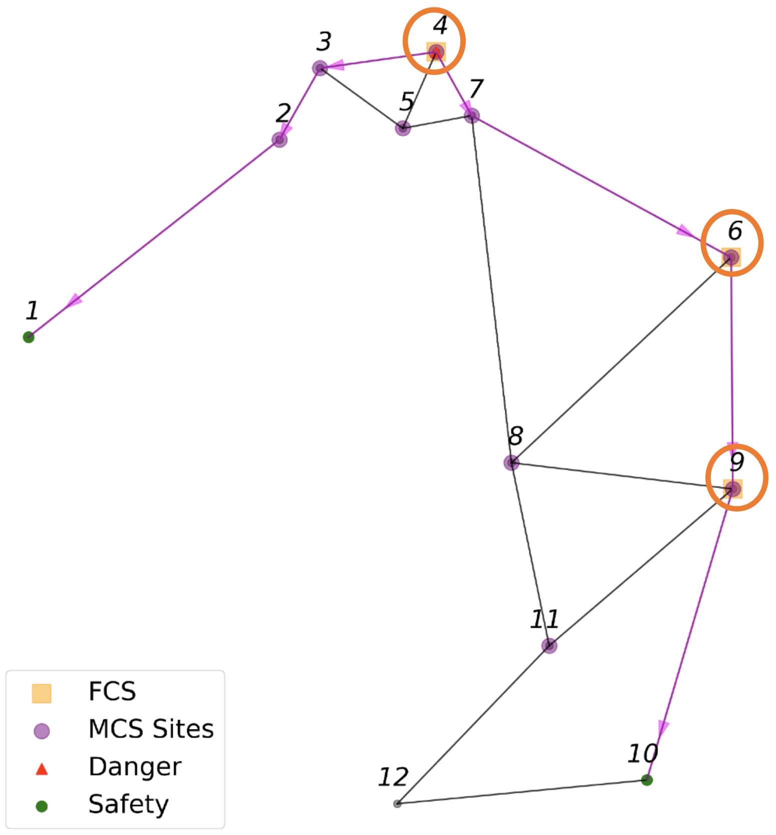
Principle for optimal scheduling: maximizing charging port occupancy

Almost 100% Occupancy

No vehicle can reach node #9 without prior charging



Example Charger Occupancy Chart with 0 MCS



Abstracted Network used in Optimization

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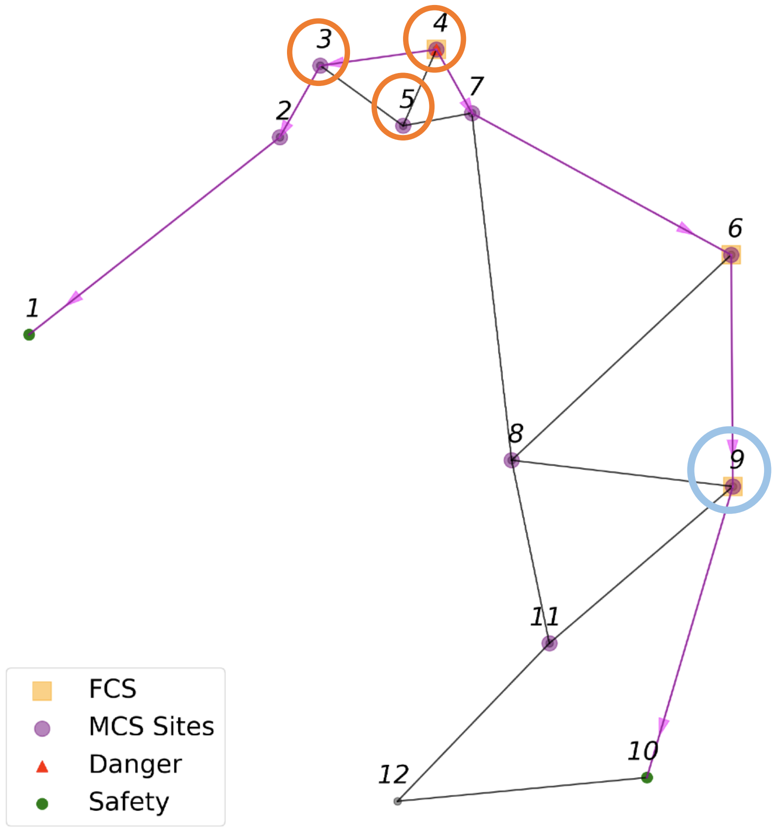
Optimal MCS Deployment Strategy



Abstracted Network used in Optimization

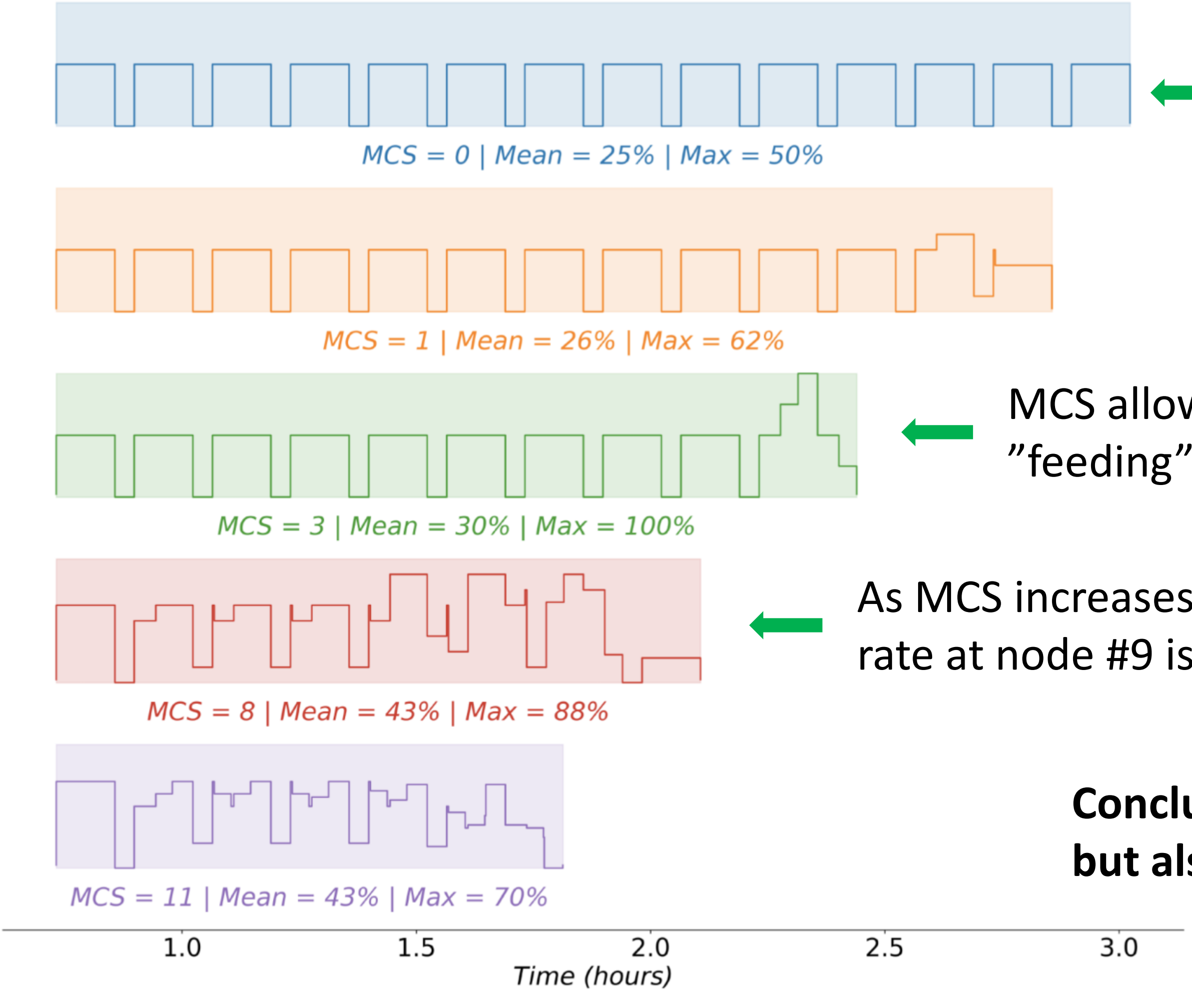
#MCS = 1 to 5:
All at node #4 ← Conjunction of shortest paths of two OD pairs

#MCS = 6 to 10:
5 at node #4
Rest at node #3 ← To supplement limited charging infrastructure for West-bound (to Merced) demand

#MCS > 10:
5 at node #4
5 at node #3
Rest at node #5 ← ”Mirror” of node #4, although not on the shortest route

Node #9 has 40 ports
We will study how its occupancy change with number of MCS

Increasing utilization of existing charging infrastructure at Node #9



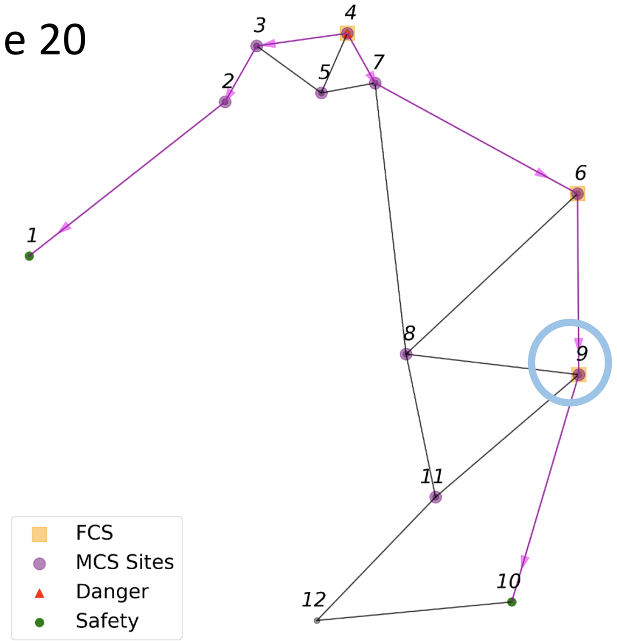
Charger Occupancy Log at Node #9

No vehicle can reach node #9 without charging first

No MCS, every fleet is limited by the 20 ports at node #4 and node #6

MCS allowing more charging near origin, ”feeding” more vehicles to node #9

As MCS increases, a growing occupancy rate at node #9 is noticed.



Node #9 in the Abstract Mariposa Map

Conclusions: MCSs not only provide direct charging opportunities, but also increase the utilization of existing charging infrastructure!

Compound Effect!

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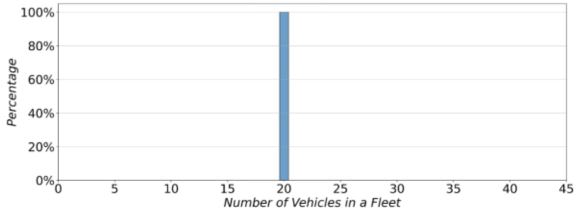
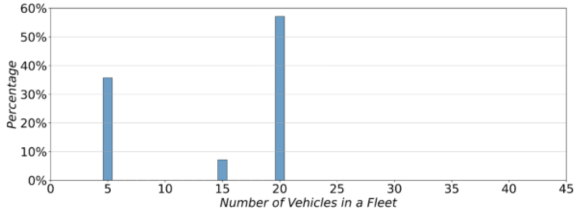
Optimal Vehicle Fleet Sizes: correlation with port number

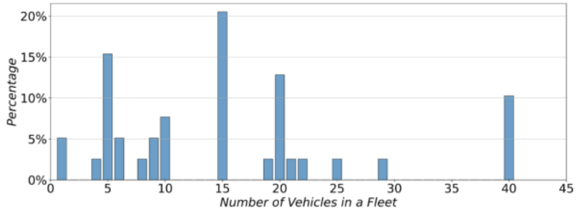
Vehicle grouping: number of vehicles taking the same evacuation schedule;



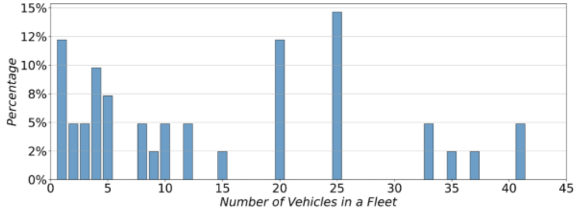
(a) Distribution without MCS



(b) Distribution with 1 MCS



(c) Distribution with 8 MCSs



(d) Distribution with 11 MCSs

Fleet Size Distributions, Homogeneous, 0, 1, 8, and 11 MCSs

MCS = 0: all groups with 20 vehicles,
which are the FCS ports at node #4 and node #6

MCS = 1: linear combo of FCS ports (20) and MCS ports (5)

MCS >8: more diverse group size as the charging capabilities are being shared between routes (interleaved), but correlation is still there.

Underlying reason: maximizing the port occupancy rate!

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Recap of Findings

MCS can significantly accelerate evacuation operation by

- Providing **direct** charging capabilities to the system
- Bringing the **compound effect** of allowing more vehicles to use further (previously **out-of-range** charging infrastructures)

An organized evacuation schedule usually consists of :

- The **shortest routes** with **charging** capabilities
- The **staggered** departure **schedule** and designed **group size** to **maximize the charging port usage**

Optimal MCS placement strategy often is:

- Near the **conjunctions** of the **shortest evacuation routes**
- **Favoring** the evacuation demand that has **limited** charging capacity in the first place

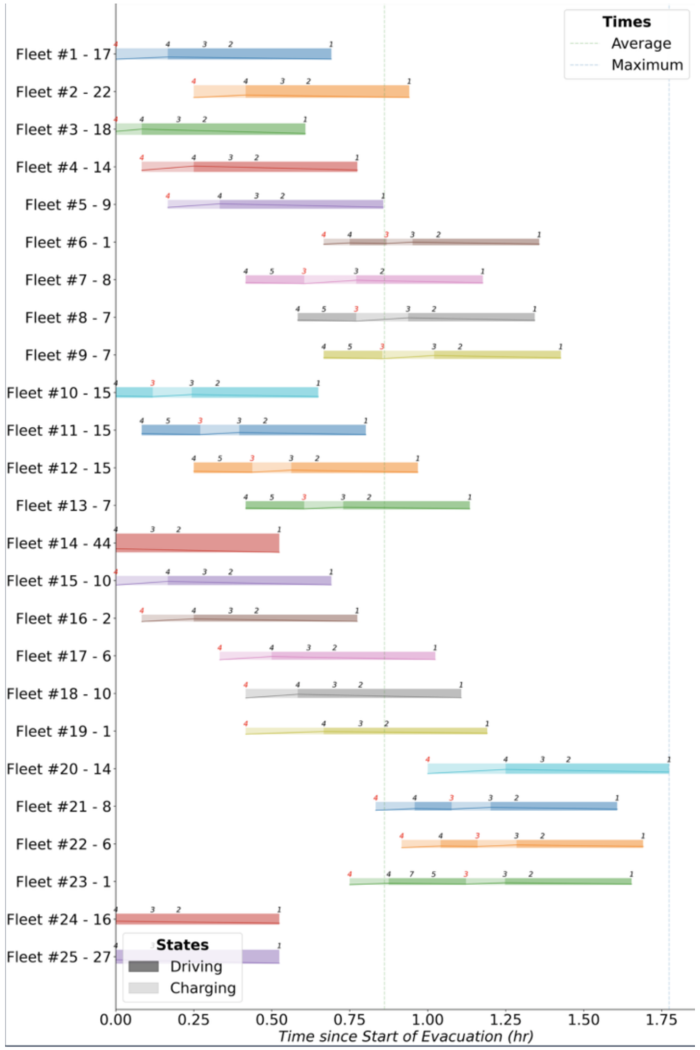
Backup: Handling Heterogeneous Demand

Case	Battery Size (kWh)	Initial SOC
Homogeneous	30 (Fixed)	0.2 (Fixed)
Heterogeneous	Uniform $\mathcal{U}\{20, 30, 40\}$	Truncated Normal $\mathcal{N}_T(0.2, 0.1, [0.0, 1.0])$

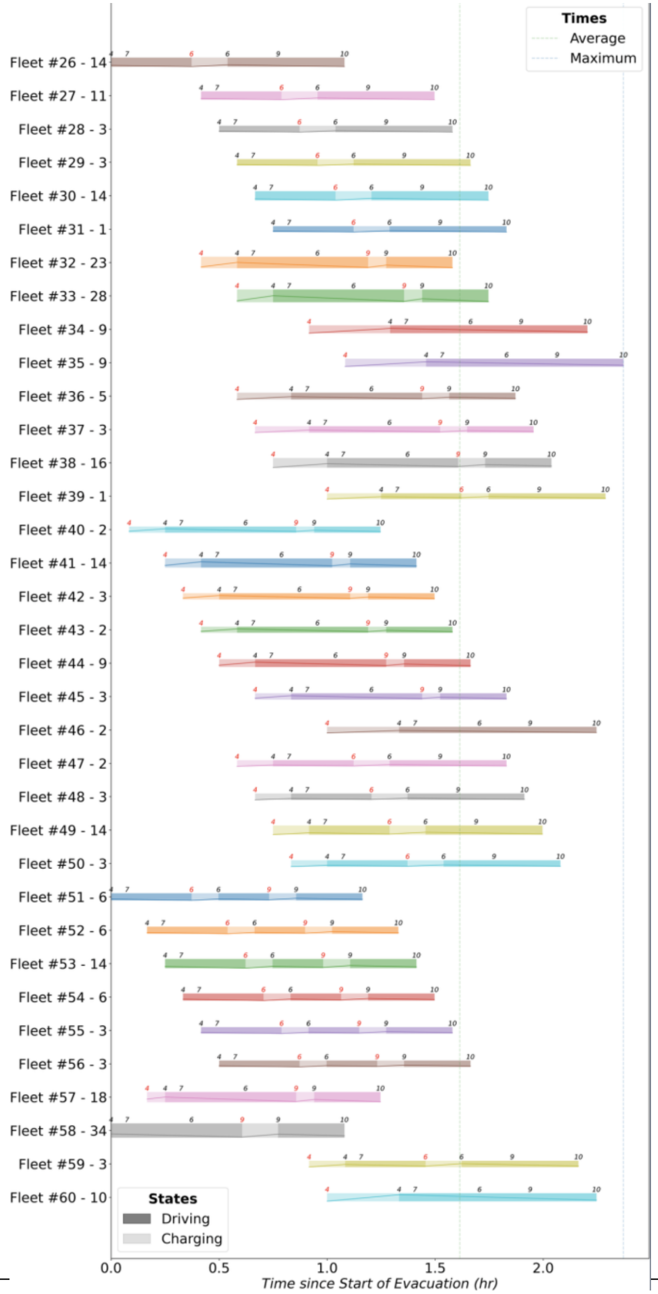
Parameters for Heterogeneous Demand

Most of the points **still stand**

Vehicle grouping will be **more diverse**



Evacuation Progress for West-bound Traffic



Evacuation Progress for South-bound Traffic

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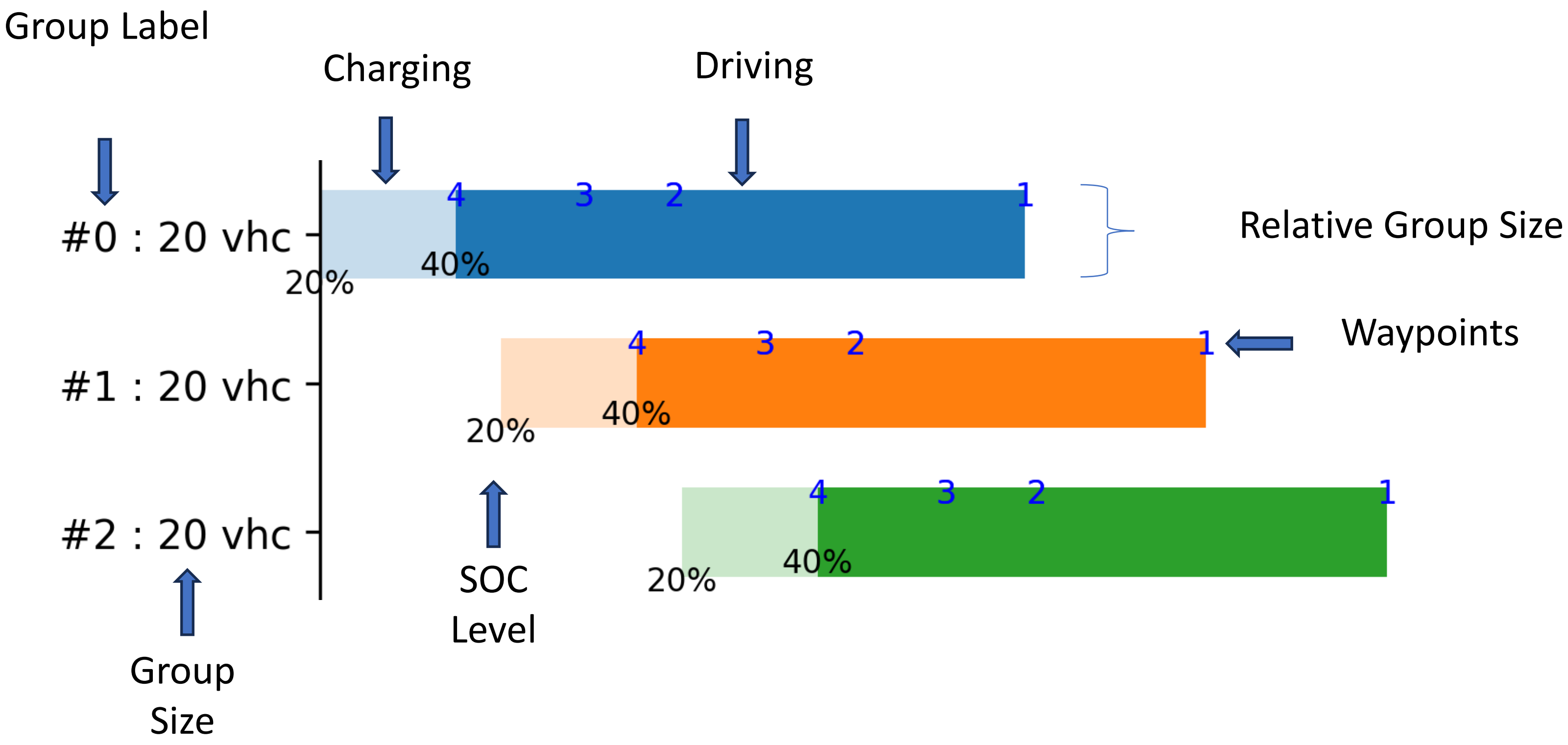
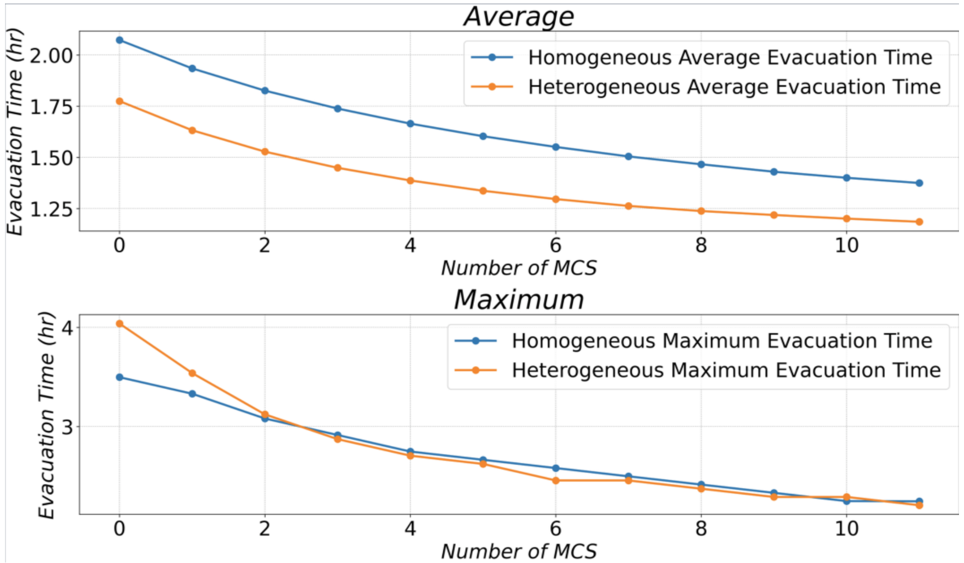
Backup: Gantt Chart Bars



Backup: Wait Time Chart

Evacuation time: Start of evacuation → Arrival of the vehicle

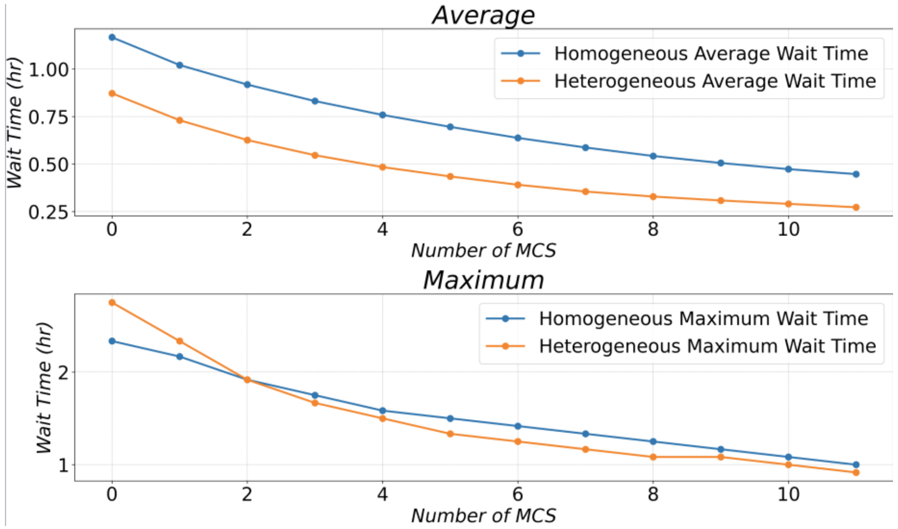


MCS Reduces **Evacuation** Time

	1 MCS	4 MCS
Homogeneous	7%	20%
Heterogeneous	8%	22%

Percentage Reduction in Average **Evacuation** Time

Wait time: Start of evacuation → Departure of the vehicle



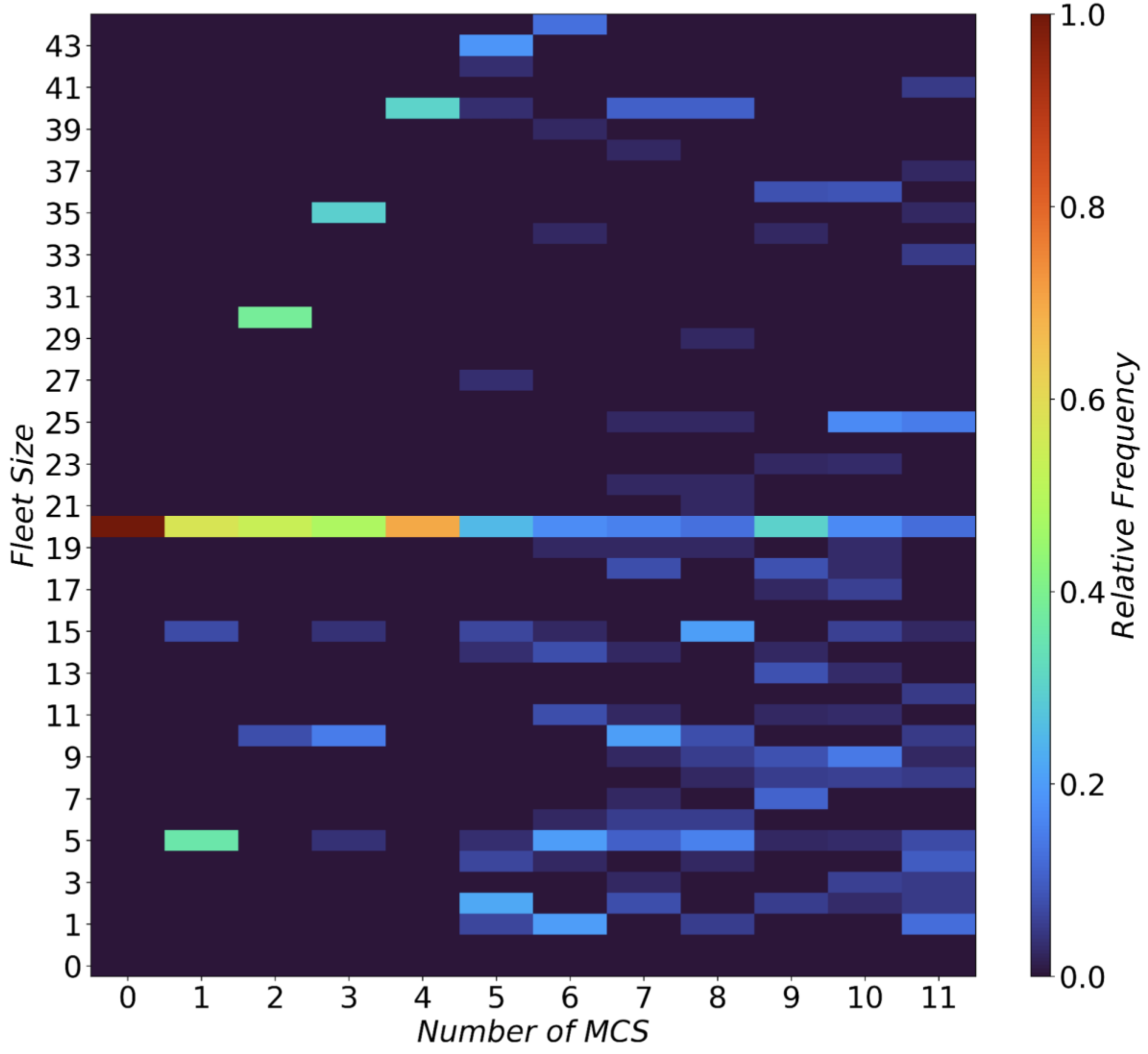
MCS Reduces **Wait** Time

	1 MCS	4 MCS
Homogeneous	14%	35%
Heterogeneous	16%	45%

Percentage Reduction in Average **Wait** Time

With 25% increase in charging infrastructure (4 MCS)

33% reduction in **maximum evacuation time** & 45% reduction in **average wait time**



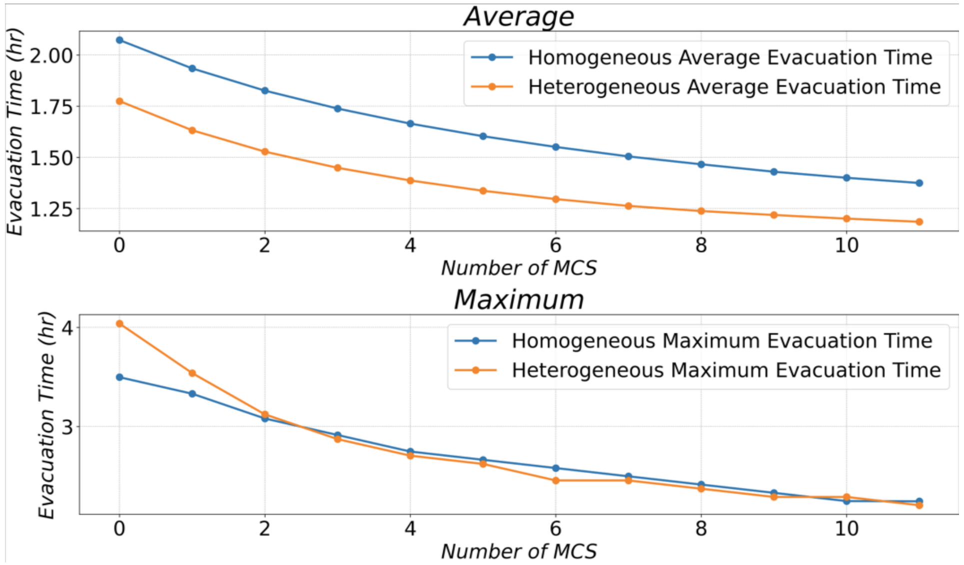
Heatmap of Fleet Size Distribution

The peaks are correlated with number of ports

The intensity of the bright band diminishes with more MCS

MCS Accelerates Evacuation

Evacuation time: Start of evacuation → Arrival of the vehicle



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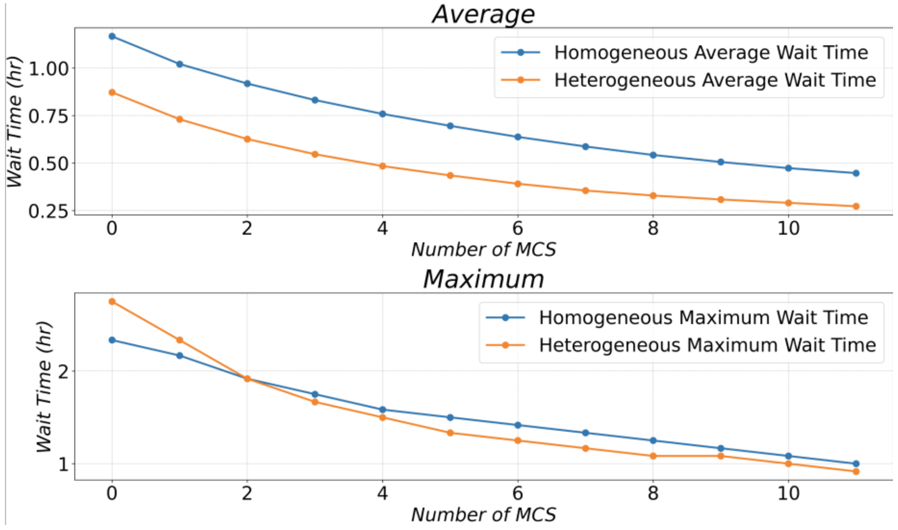

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Thank you!